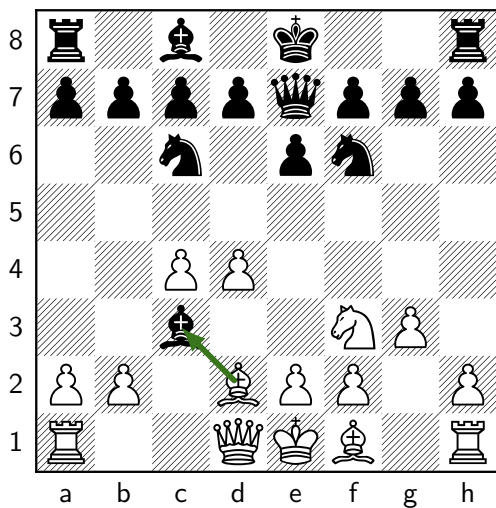


Melanie-ANALOGY — Bogo-Indian

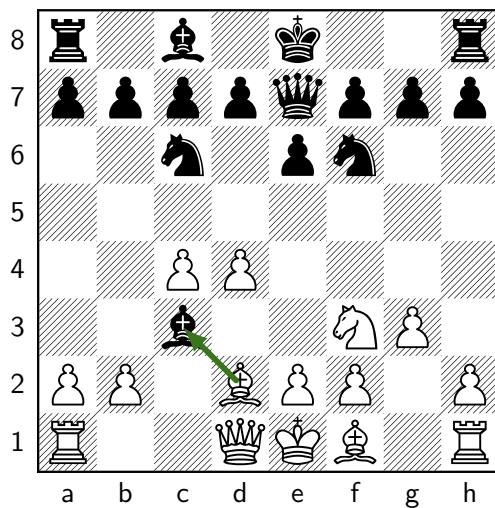
Melanie (Deep_Vision)

Abstract

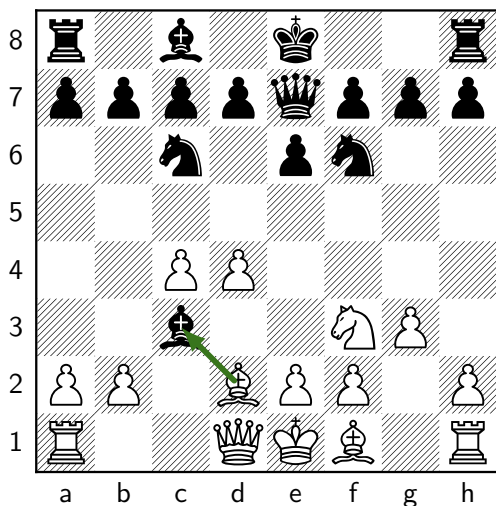
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



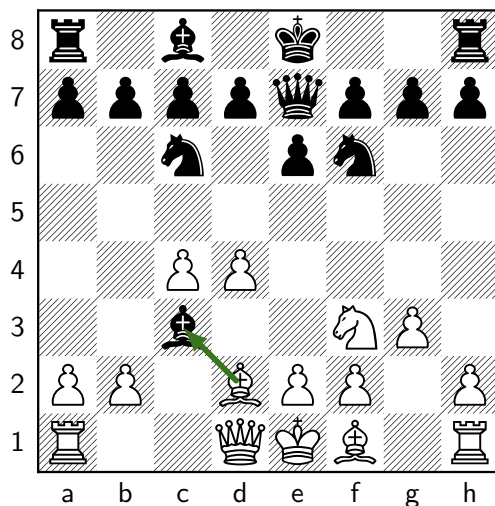
(a)



(b)

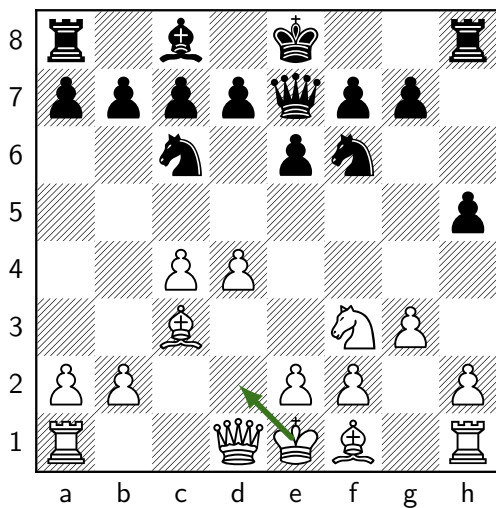


(c)

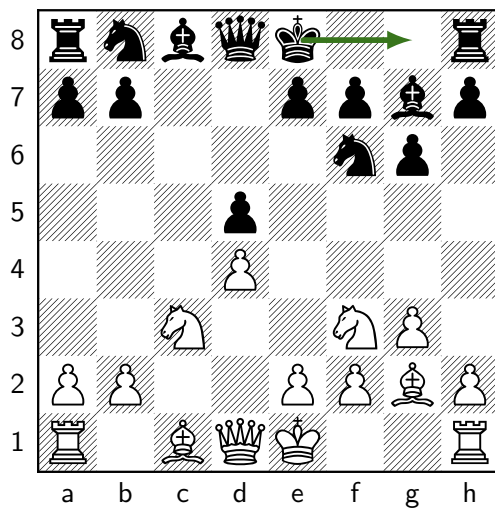


(d)

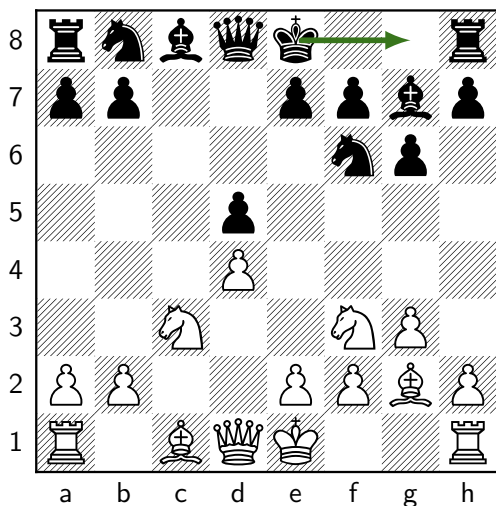
Figure 1: **Move 7, White: Bxc3.** (a) Our position; Melanie plays Bxc3 (d2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#3642` (ply 12) [structural distance $d=0$, QAP obj=193]: the master played Bxc3; under μ our Bd2 plays the role of that game’s Bd2; cross-type roles: our Pe2 as their Bf1, our ra8 as their Pe2; transferred move d2c3 (**endorses this move**). (c) Retrieved from `cctl_top_p0000#3643` (ply 12) [structural distance $d=0$, QAP obj=193]: the master played Bxc3; under μ our Bd2 plays the role of that game’s Bd2; cross-type roles: our Pe2 as their Bf1, our ra8 as their Pe2; transferred move d2c3 (**endorses this move**). (d) Retrieved from `cctl_top_p0002#1512` (ply 12) [structural distance $d=0$, QAP obj=193]: the master played Bxc3; under μ our Bd2 plays the role of that game’s Bd2; cross-type roles: our Pe2 as their Bf1, our ra8 as their Pe2; transferred move d2c3 (**endorses this move**). Synthesis: 47 retrieved analogs transferred to Bxc3 (summed QAP objective 6866, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



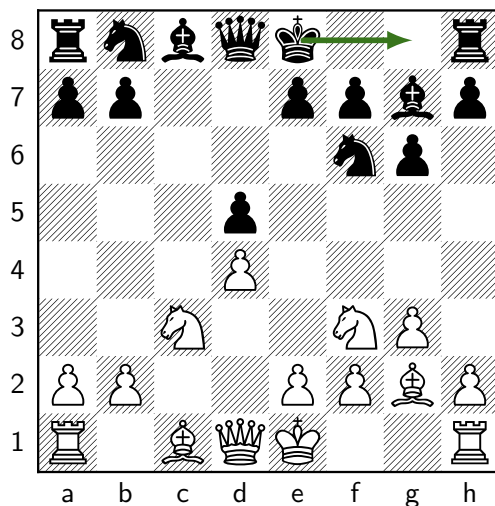
(a)



(b)

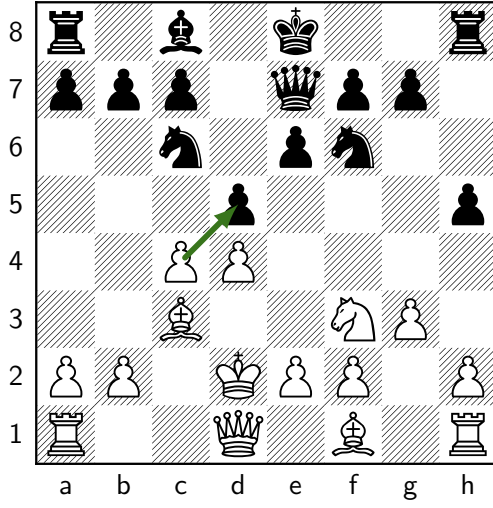


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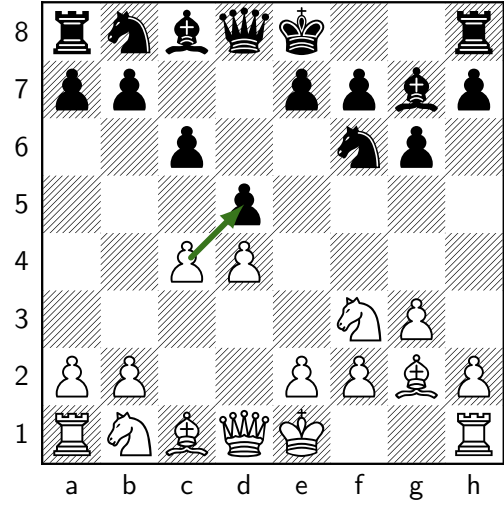


(d)

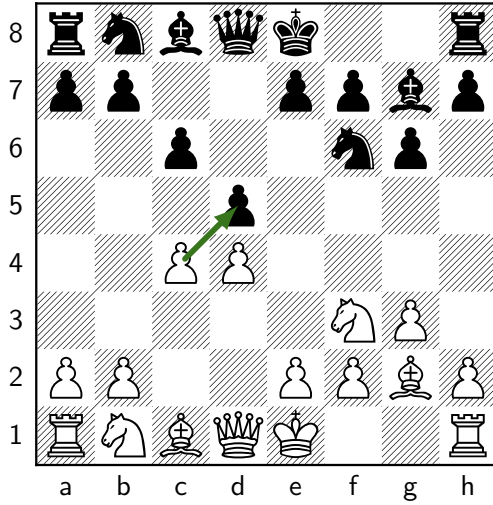
Figure 2: **Move 8, White: Kd2.** (a) Our position; Melanie plays Kd2 (e1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2563` (ply 13) [structural distance $d=59$, QAP obj=35]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bc8* as their *Ra8*, our *pg7* as their *Nb8*; transferred move e1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2564` (ply 13) [structural distance $d=59$, QAP obj=35]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bc8* as their *Ra8*, our *pg7* as their *Nb8*; transferred move e1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#753` (ply 13) [structural distance $d=59$, QAP obj=35]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bc8* as their *Ra8*, our *pg7* as their *Nb8*; transferred move e1d2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kd2 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



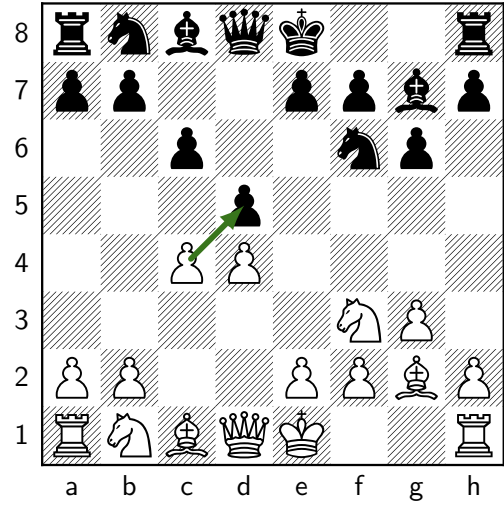
(a)



(b)

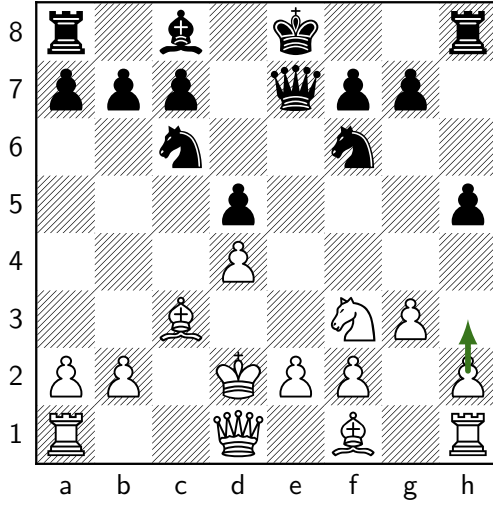


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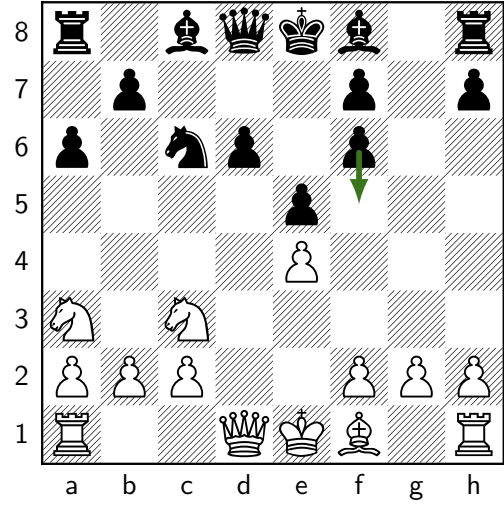


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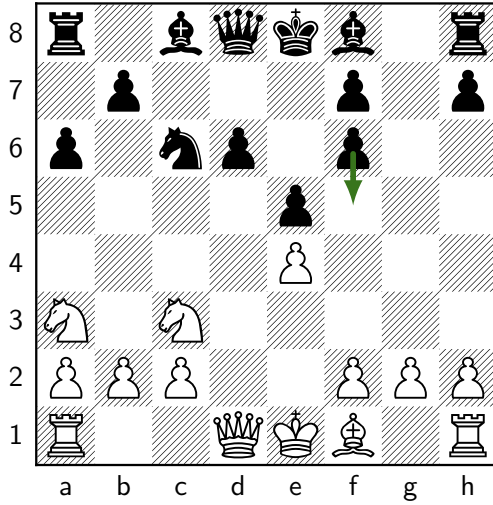
Figure 3: **Move 9, White: cxd5.** (a) Our position; Melanie plays cxd5 (c4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#878 (ply 10) [structural distance $d=60$, QAP obj=46]: the master played cxd5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Bf1$ as their $Ra1$, our $Qd1$ as their $Nb1$; transferred move c4d5 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#879 (ply 10) [structural distance $d=60$, QAP obj=46]: the master played cxd5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Bf1$ as their $Ra1$, our $Qd1$ as their $Nb1$; transferred move c4d5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#88 (ply 10) [structural distance $d=60$, QAP obj=46]: the master played cxd5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Bf1$ as their $Ra1$, our $Qd1$ as their $Nb1$; transferred move c4d5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to cxd5 (summed QAP objective 229, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



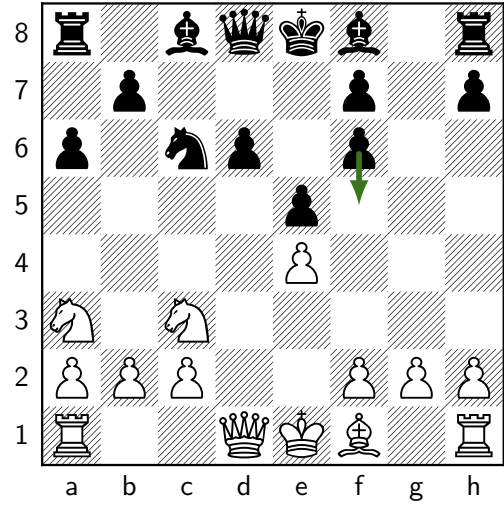
(a)



(b)

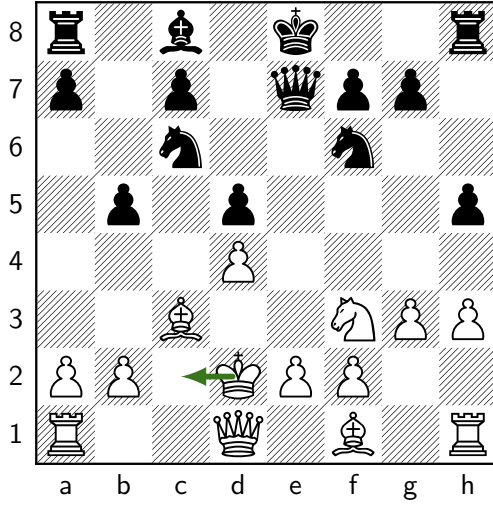


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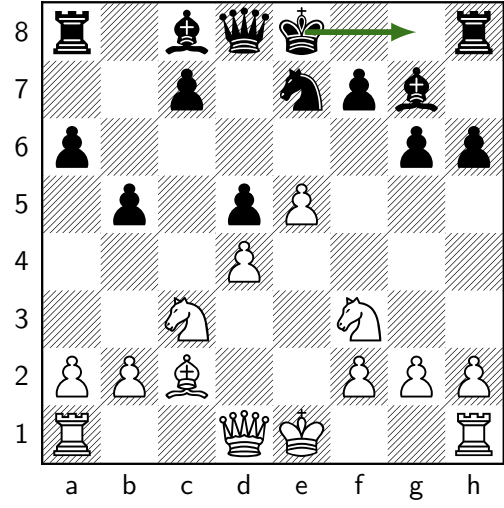


(d)

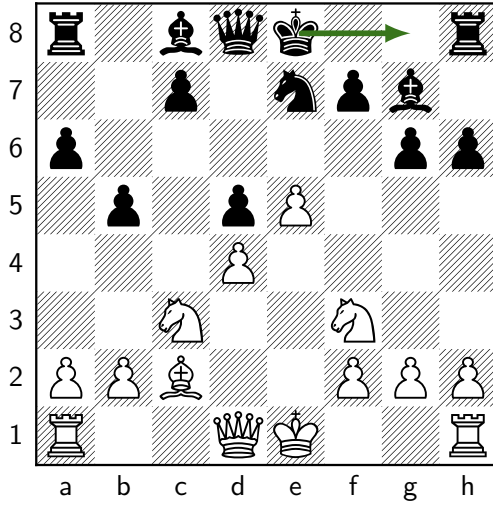
Figure 4: **Move 10, White: h3.** (a) Our position; Melanie plays **h3** (**h2h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#269` (ply 17) [structural distance $d=60$, QAP obj=26]: the master played **f5**; under μ our *Ph2* plays the role of that game’s *Pf3*; cross-type roles: our *nf6* as their *Ra8*, our *ph5* as their *Bc8*; transferred move **h2h3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#270` (ply 17) [structural distance $d=60$, QAP obj=26]: the master played **f5**; under μ our *Ph2* plays the role of that game’s *Pf3*; cross-type roles: our *nf6* as their *Ra8*, our *ph5* as their *Bc8*; transferred move **h2h3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1650` (ply 17) [structural distance $d=60$, QAP obj=26]: the master played **f5**; under μ our *Ph2* plays the role of that game’s *Pf3*; cross-type roles: our *nf6* as their *Ra8*, our *ph5* as their *Bc8*; transferred move **h2h3** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **h3** (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



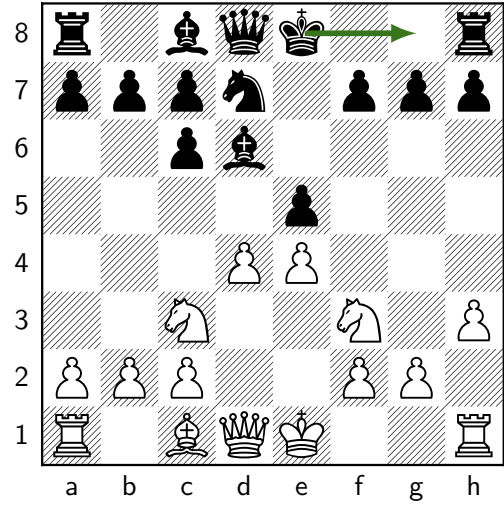
(a)



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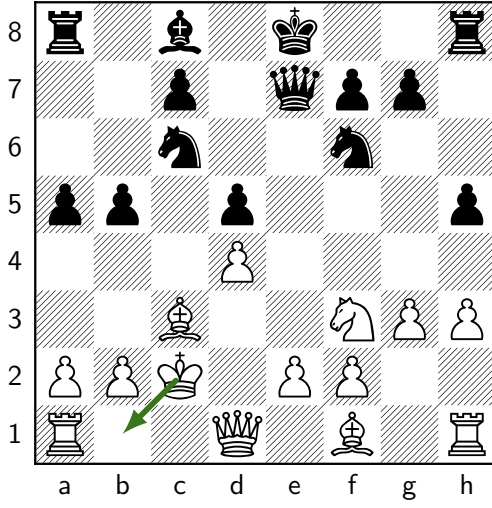


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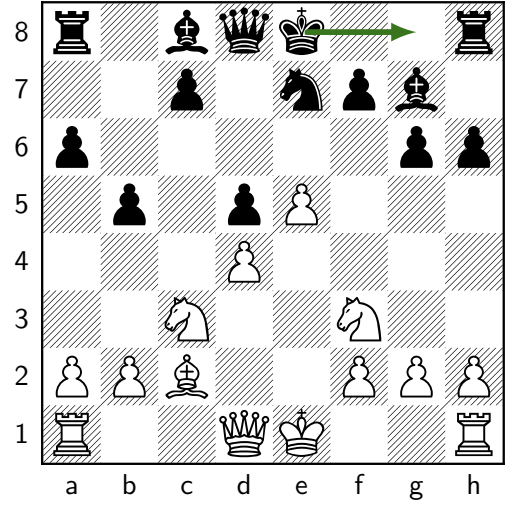


(d)

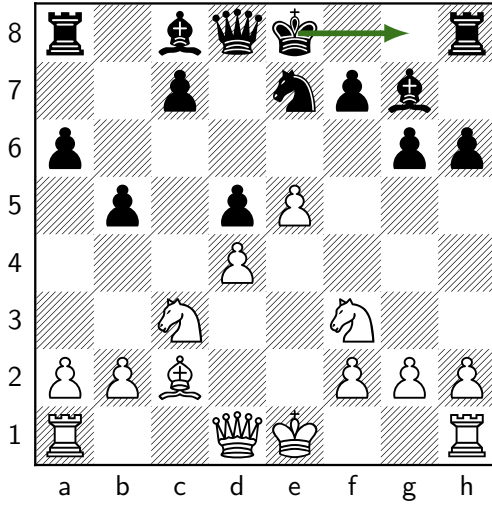
Figure 5: **Move 11, White: Kc2.** (a) Our position; Melanie plays Kc2 (d2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1943` (ply 23) [structural distance $d=64$, QAP obj=55]: the master played 0-0; under μ our *Kd2* plays the role of that game's *Ke1*; cross-type roles: our *Bc3* as their *Ra8*, our *Pg3* as their *Bc8*; transferred move d2c2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2098` (ply 23) [structural distance $d=64$, QAP obj=55]: the master played 0-0; under μ our *Kd2* plays the role of that game's *Ke1*; cross-type roles: our *Bc3* as their *Ra8*, our *Pg3* as their *Bc8*; transferred move d2c2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1404` (ply 13) [structural distance $d=70$, QAP obj=19]: the master played 0-0; under μ our *Kd2* plays the role of that game's *Ke1*; cross-type roles: our *Bf1* as their *Ra8*, our *pd5* as their *Bc8*; transferred move d2c2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kc2 (summed QAP objective 138, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



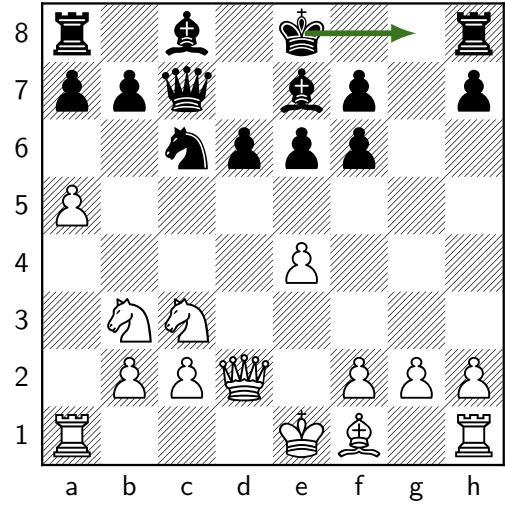
(a)



(b)

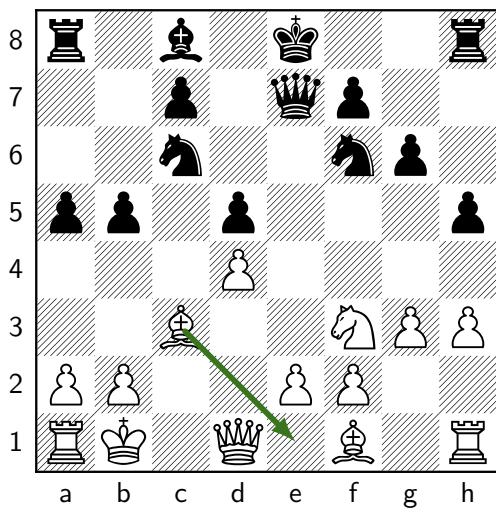


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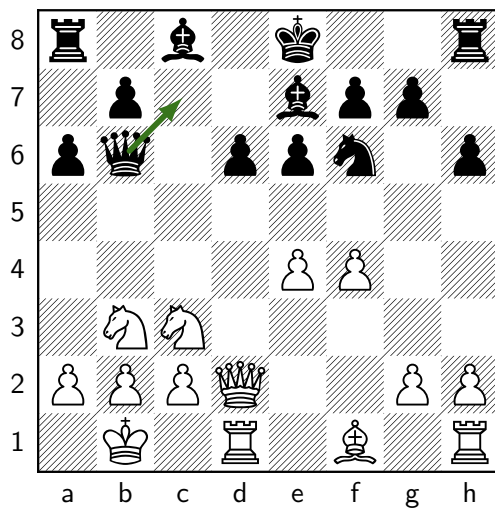


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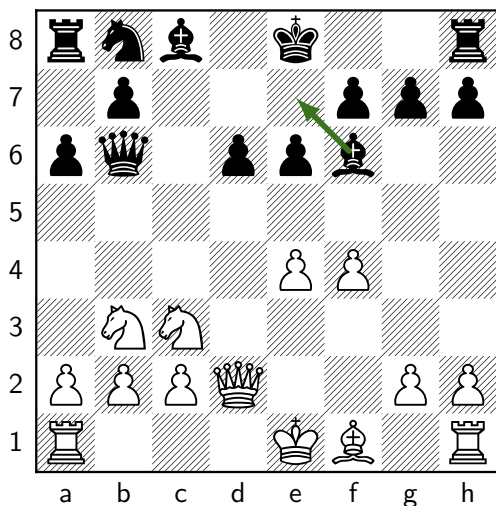
Figure 6: **Move 12, White: Kb1.** (a) Our position; Melanie plays Kb1 (c2b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1943` (ply 23) [structural distance $d=56$, QAP obj=70]: the master played 0-0; under μ our *Kc2* plays the role of that game’s *Ke1*; cross-type roles: our *Pg3* as their *Bc8*, our *Ra1* as their *Qd8*; transferred move c2b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2098` (ply 23) [structural distance $d=56$, QAP obj=70]: the master played 0-0; under μ our *Kc2* plays the role of that game’s *Ke1*; cross-type roles: our *Pg3* as their *Bc8*, our *Ra1* as their *Qd8*; transferred move c2b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3849` (ply 21) [structural distance $d=57$, QAP obj=51]: the master played 0-0; under μ our *Kc2* plays the role of that game’s *Ke1*; cross-type roles: our *Pg3* as their *Ra8*, our *Pb2* as their *Bc8*; transferred move c2b1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kb1 (summed QAP objective 192, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



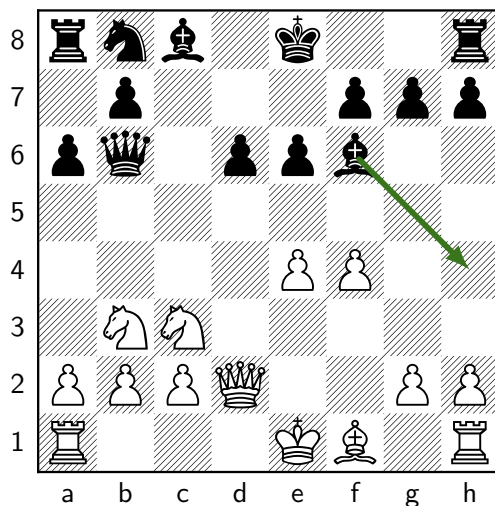
(a)



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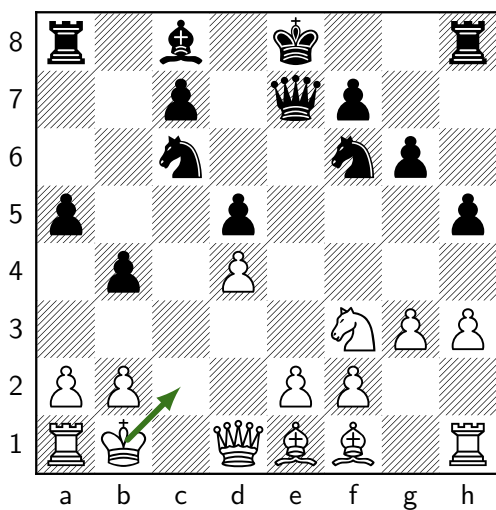


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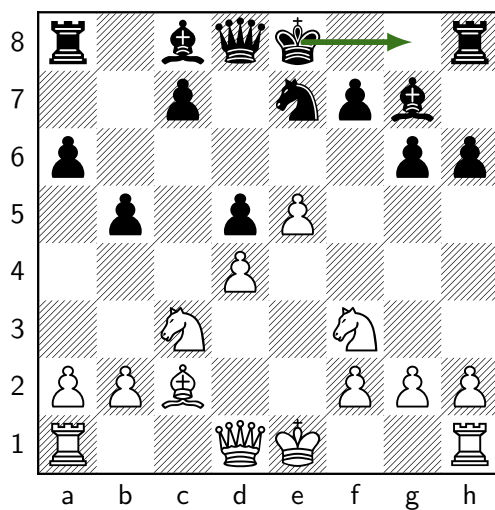


(d)

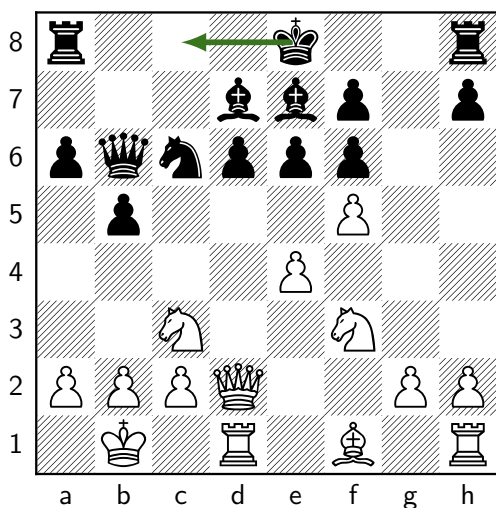
Figure 7: **Move 13, White: Be1.** (a) Our position; Melanie plays **Be1** (**c3e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#142` (ply 23) [structural distance $d=74$, QAP obj=24]: the master played **Qc7**; under μ our *Bc3* plays the role of that game’s *Qb3* (a **cross-type** correspondence); cross-type roles: our *pb5* as their *Bc8*, our *ph5* as their *Rh8*; transferred move **c3e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#505` (ply 19) [structural distance $d=73$, QAP obj=23]: the master played **Be7**; under μ our *Bc3* plays the role of that game’s *Bf3*; cross-type roles: our *pd5* as their *Ra8*, our *ph5* as their *Bc8*; transferred move **c3e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#559` (ply 19) [structural distance $d=73$, QAP obj=23]: the master played **Bh4+**; under μ our *Bc3* plays the role of that game’s *Bf3*; cross-type roles: our *pd5* as their *Ra8*, our *ph5* as their *Bc8*; transferred move **c3e1** (**endorses this move**). Synthesis: 10 retrieved analogs transferred to **Be1** (summed QAP objective 235, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



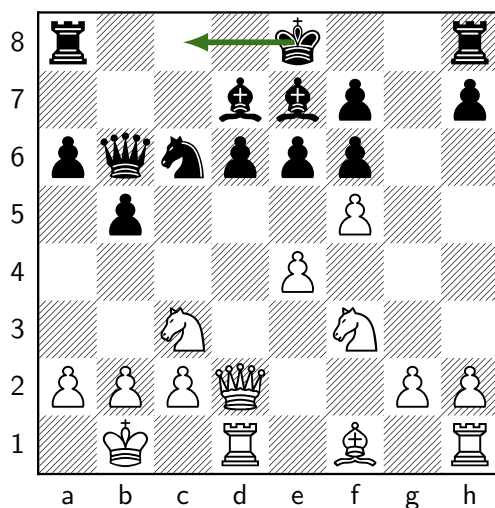
(a)



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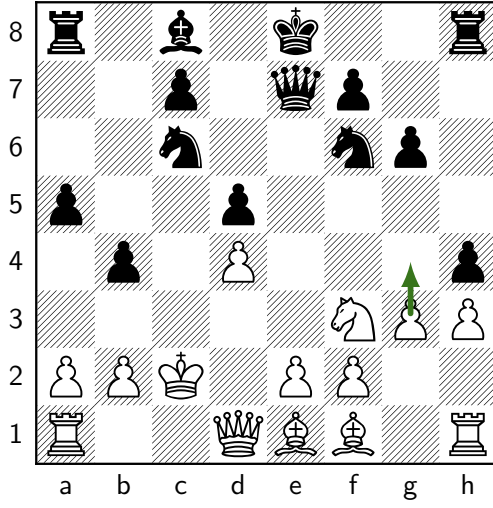


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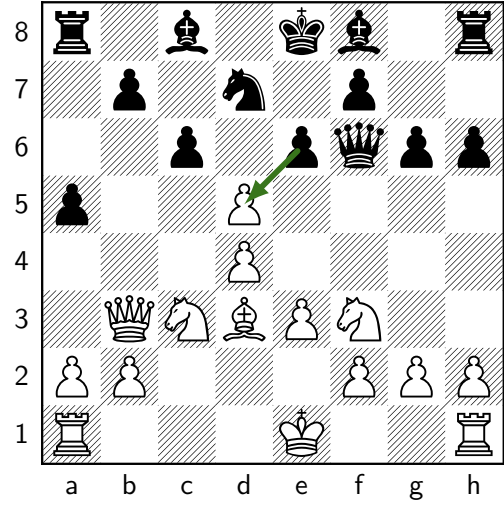


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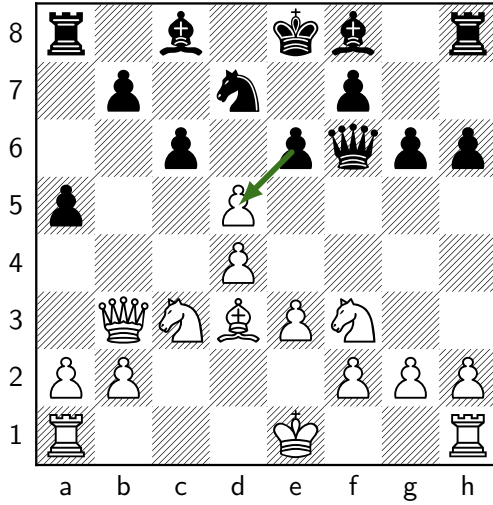
Figure 8: **Move 14, White: Kc2.** (a) Our position; Melanie plays Kc2 (b1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#1943` (ply 23) [structural distance $d=70$, QAP obj=65]: the master played 0-0; under μ our *Kb1* plays the role of that game’s *Ke1*; cross-type roles: our *ph5* as their *Bc8*, our *bc8* as their *nf3*; transferred move b1c2 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#198` (ply 25) [structural distance $d=76$, QAP obj=39]: the master played 0-0-0; under μ our *Kb1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *pg6* as their *Rh8*; transferred move b1c2 (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#2086` (ply 25) [structural distance $d=76$, QAP obj=39]: the master played 0-0-0; under μ our *Kb1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *pg6* as their *Rh8*; transferred move b1c2 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Kc2 (summed QAP objective 486, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



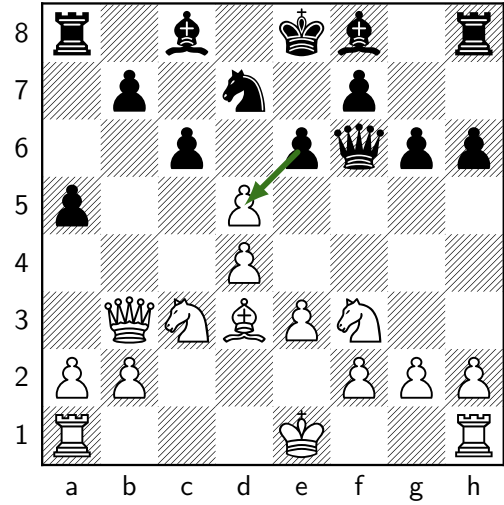
(a)



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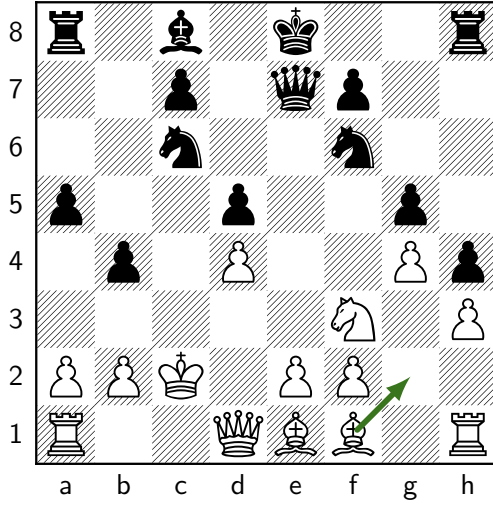


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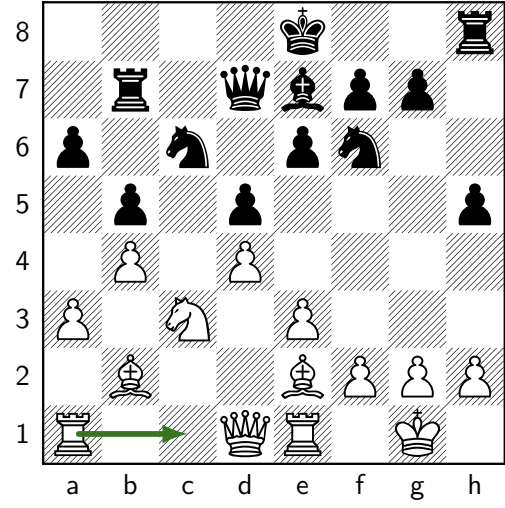


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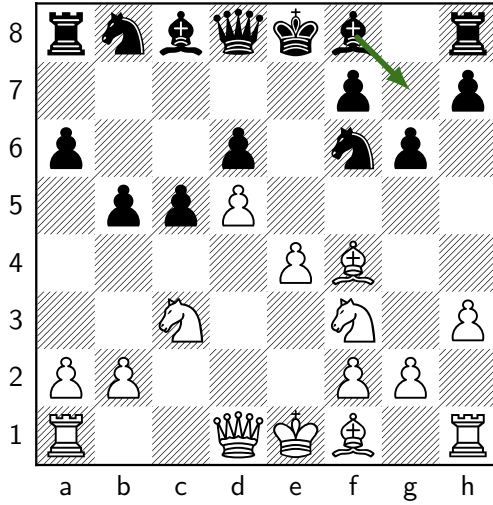
Figure 9: **Move 15, White: g4 (g3g4).** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3214` (ply 19) [structural distance $d=81$, QAP obj=94]: the master played `exd5`; under μ our $Pg3$ plays the role of that game’s $Pe3$; cross-type roles: our $pa5$ as their $Ra8$, our $pc7$ as their $Bc8$; transferred move `g3g4` (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2056` (ply 19) [structural distance $d=81$, QAP obj=94]: the master played `exd5`; under μ our $Pg3$ plays the role of that game’s $Pe3$; cross-type roles: our $pa5$ as their $Ra8$, our $pc7$ as their $Bc8$; transferred move `g3g4` (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2057` (ply 19) [structural distance $d=81$, QAP obj=94]: the master played `exd5`; under μ our $Pg3$ plays the role of that game’s $Pe3$; cross-type roles: our $pa5$ as their $Ra8$, our $pc7$ as their $Bc8$; transferred move `g3g4` (**endorses this move**). Synthesis: 5 retrieved analogs transferred to g4 (summed QAP objective 371, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



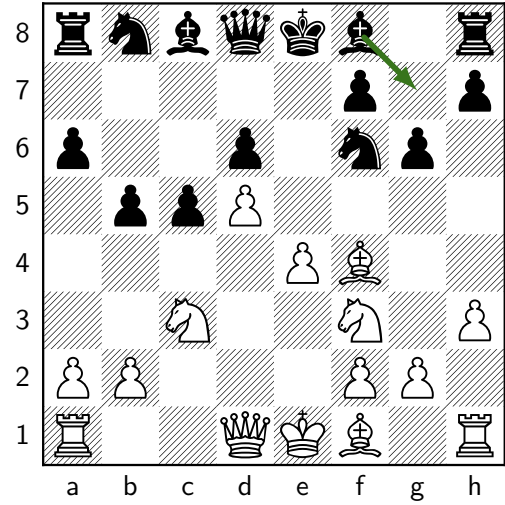
(a)



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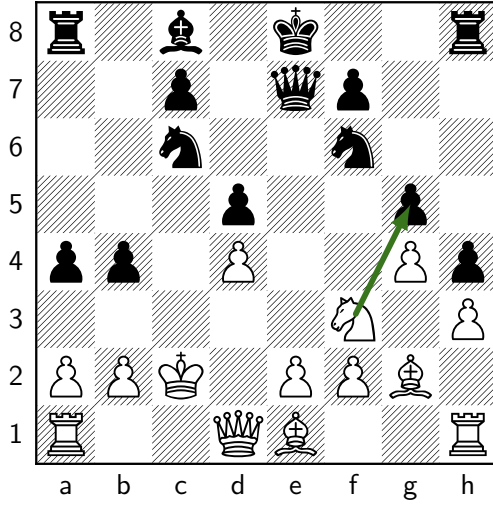


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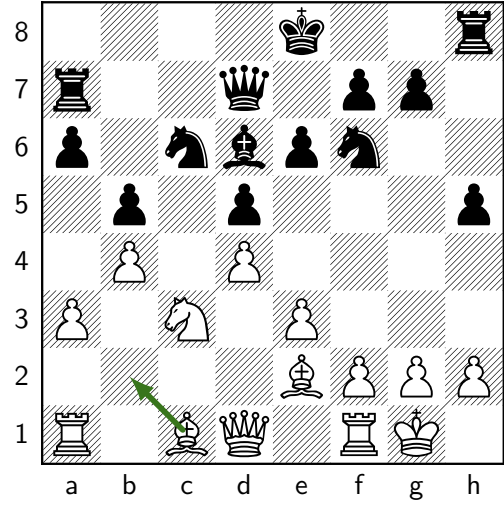


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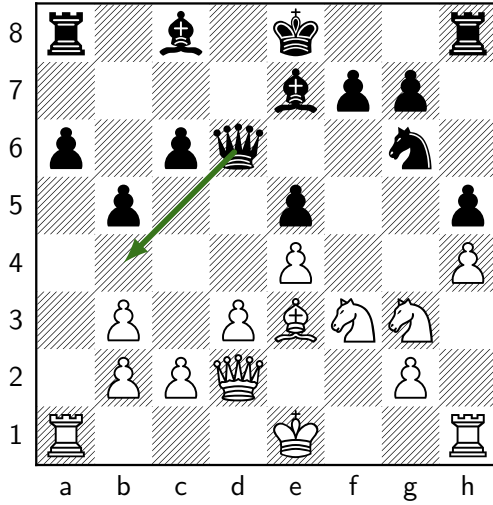
Figure 10: **Move 16, White: Bg2.** (a) Our position; Melanie plays Bg2 (f1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#762 (ply 32) [structural distance $d=82$, QAP obj=119]: the master played Rc1; under μ our Bf1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Qd1, our Qd1 as their Re1; transferred move f1g2 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#764 (ply 17) [structural distance $d=85$, QAP obj=109]: the master played Bg7; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pa2 as their Ra8, our Pf2 as their Nb8; transferred move f1g2 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#765 (ply 17) [structural distance $d=85$, QAP obj=109]: the master played Bg7; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pa2 as their Ra8, our Pf2 as their Nb8; transferred move f1g2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bg2 (summed QAP objective 554, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



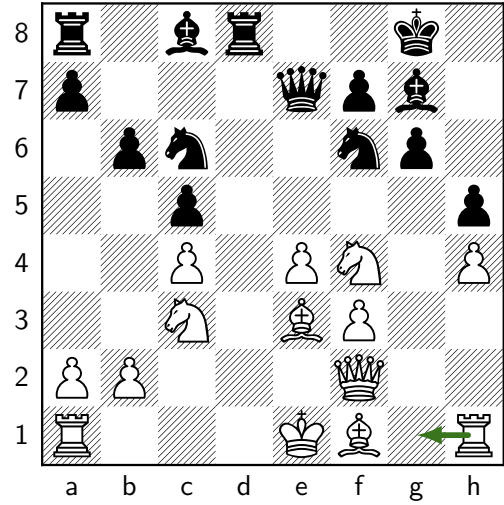
(a)



(b)

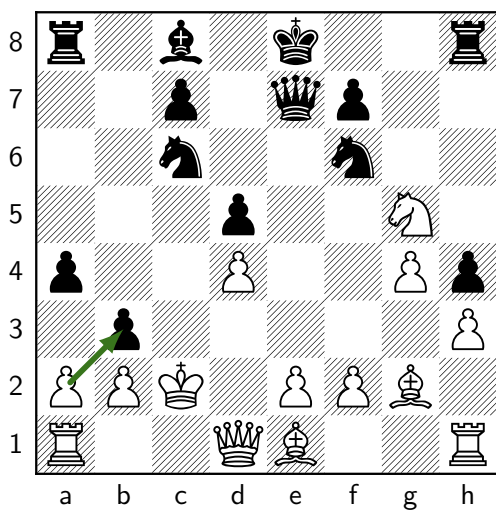


(c)

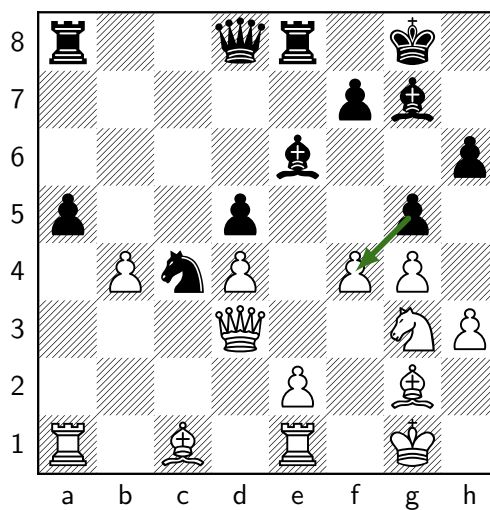


(d)

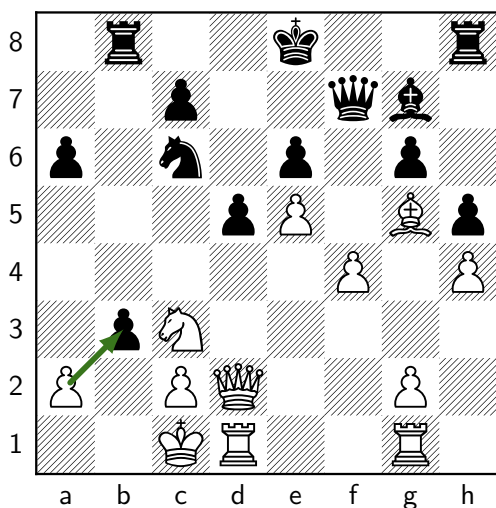
Figure 11: **Move 17, White: Nxg5.** (a) Our position; Melanie plays Nxg5 (f3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#762` (ply 28) [structural distance $d=81$, QAP obj=130]: the master played Bb2; under μ our Ra1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our pd5 as their Ra1, our Bg2 as their Qd1; transferred move a1b1. (c) Retrieved from `ccr1_top_p0000#2010` (ply 29) [structural distance $d=75$, QAP obj=106]: the master played Qb4; under μ our Be1 plays the role of that game’s Qd3 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Ra8, our ph4 as their Bc8; transferred move e1b4. (d) Retrieved from `ccr1_top_p0001#3297` (ply 30) [structural distance $d=87$, QAP obj=105]: the master played Rg1; under μ our Bg2 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Ra1, our pa4 as their Bf1; transferred move g2f1. Synthesis: no retrieved analog transferred to Nxg5 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



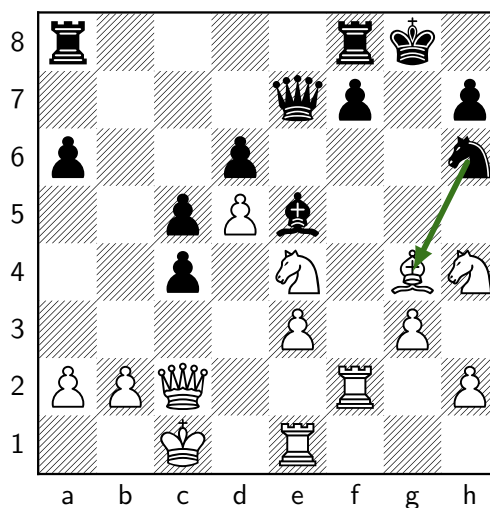
(a)



(b)

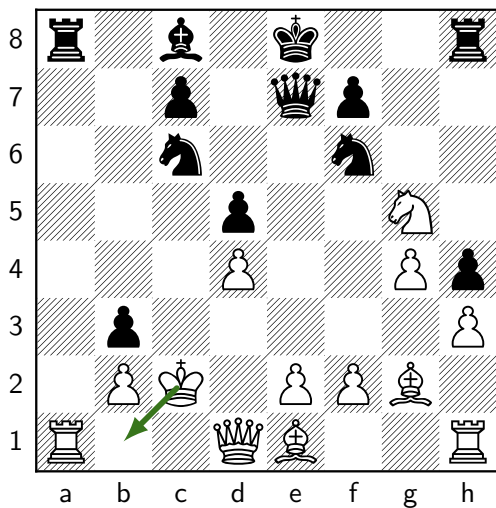


(c)

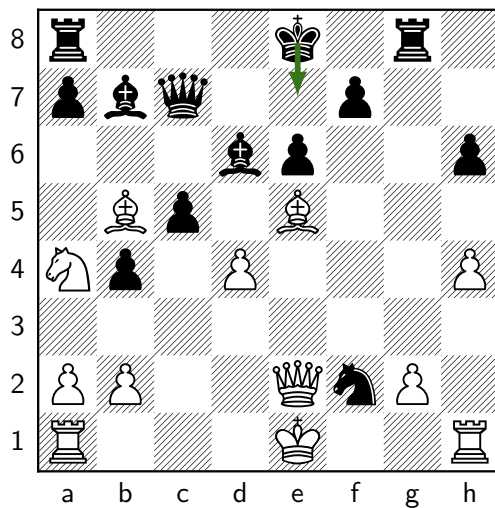


(d)

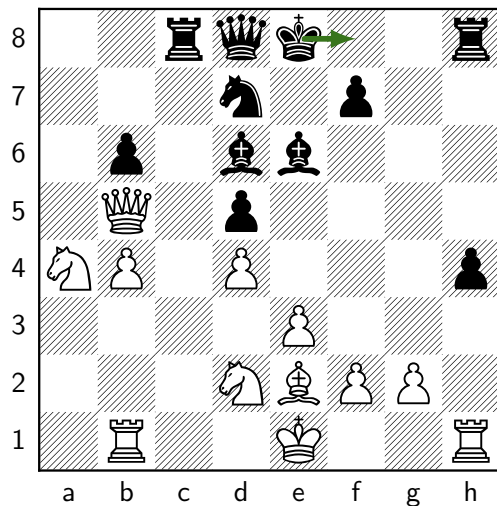
Figure 12: **Move 18, White: axb3.** (a) Our position; Melanie plays axb3 (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#2670 (ply 51) [structural distance $d=109$, QAP obj=52]: the master played gxf4; under μ our Pa2 plays the role of that game’s Pg4; cross-type roles: our Be1 as their Ra8, our nf6 as their Be6; transferred move a2b3 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2040 (ply 40) [structural distance $d=107$, QAP obj=26]: the master played axb3; under μ our Pa2 plays the role of that game’s Pa2; cross-type roles: our Qd1 as their Rd1, our Pe2 as their Rg1; transferred move a2b3 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#638 (ply 45) [structural distance $d=107$, QAP obj=23]: the master played Nxg4; under μ our Pa2 plays the role of that game’s Nh3 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Rf8, our Be1 as their Qe7; transferred move a2b3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to axb3 (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

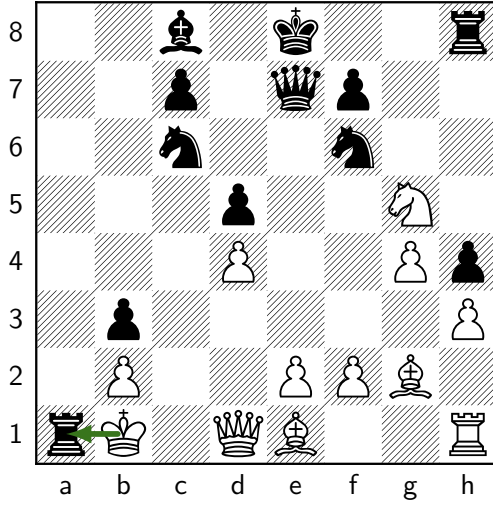


(b)

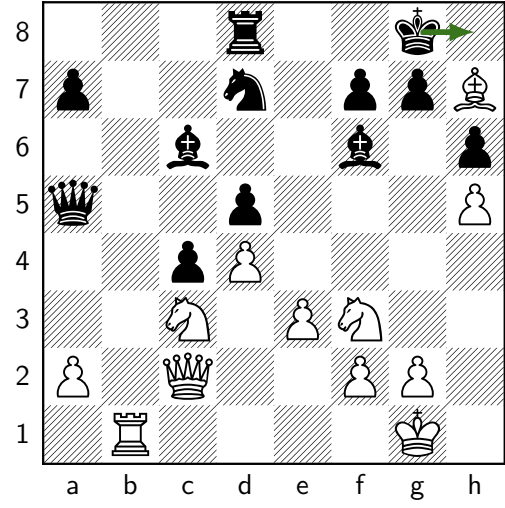


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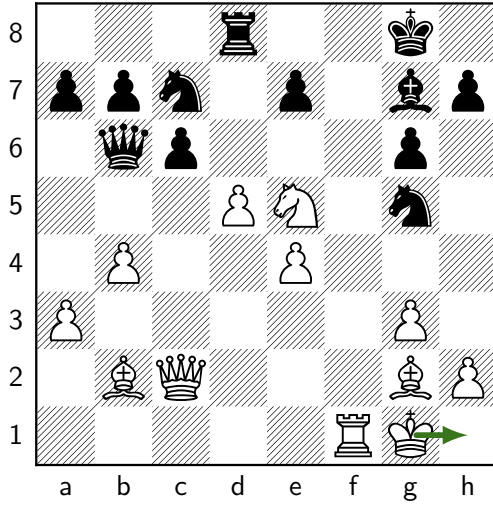
Figure 13: **Move 19, White: Kb1.** (a) Our position; Melanie plays Kb1 (c2b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2130` (ply 39) [structural distance $d=106$, QAP obj=31]: the master played Ke7; under μ our Kc2 plays the role of that game’s Ke1; cross-type roles: our Qd1 as their Ra8, our Pb2 as their Rg8; transferred move c2d2. (c) Retrieved from `ccr1_top_p0002#3091` (ply 45) [structural distance $d=116$, QAP obj=12]: the master played Kf8; under μ our Kc2 plays the role of that game’s Ke1; cross-type roles: our nf6 as their Rc8, our Pg4 as their Rh8; transferred move c2c1. Synthesis: no retrieved analog transferred to Kb1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



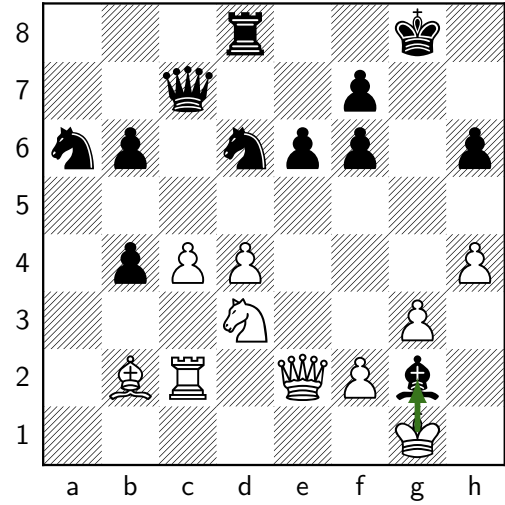
(a)



(b)

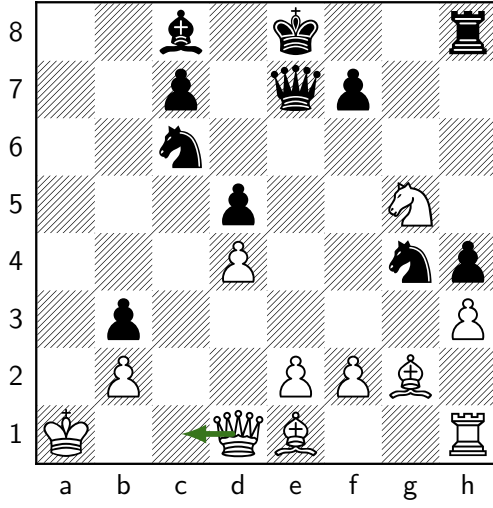


(c)

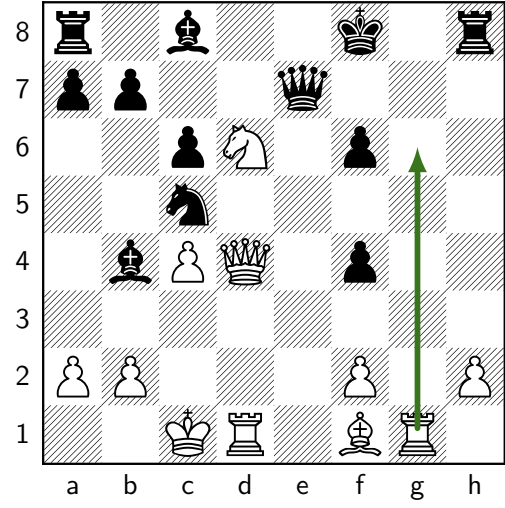


(d)

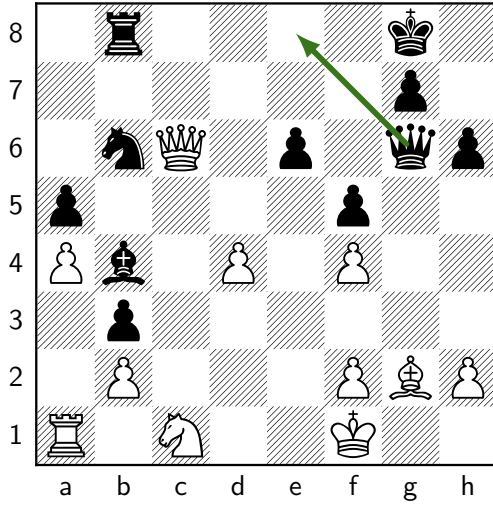
Figure 14: **Move 20, White: Kxa1.** (a) Our position; Melanie plays Kxa1 (b1a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1235` (ply 47) [structural distance $d=113$, QAP obj=86]: the master played Kh8; under μ our Kb1 plays the role of that game’s Kg1; cross-type roles: our Qd1 as their Rd8, our Be1 as their Nd7; transferred move b1a1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1087` (ply 44) [structural distance $d=104$, QAP obj=30]: the master played Kh1; under μ our Kb1 plays the role of that game’s Kg1; cross-type roles: our Qd1 as their Rf1, our Pf2 as their Bb2; transferred move b1a1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3471` (ply 50) [structural distance $d=113$, QAP obj=26]: the master played Kxg2; under μ our Kb1 plays the role of that game’s Kg1; cross-type roles: our Pf2 as their Bb2, our nf6 as their Rc2; transferred move b1a1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kxa1 (summed QAP objective 264, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



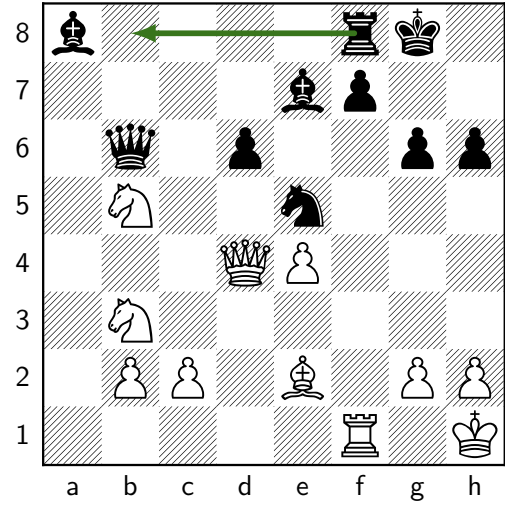
(a)



(b)

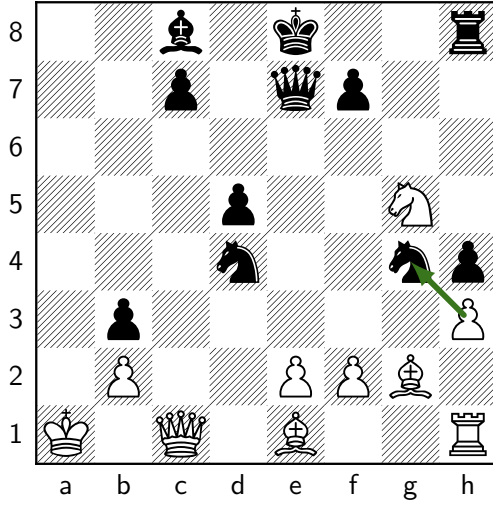


(c)

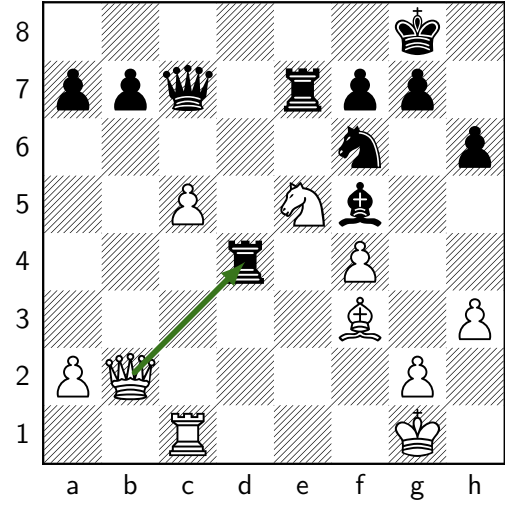


(d)

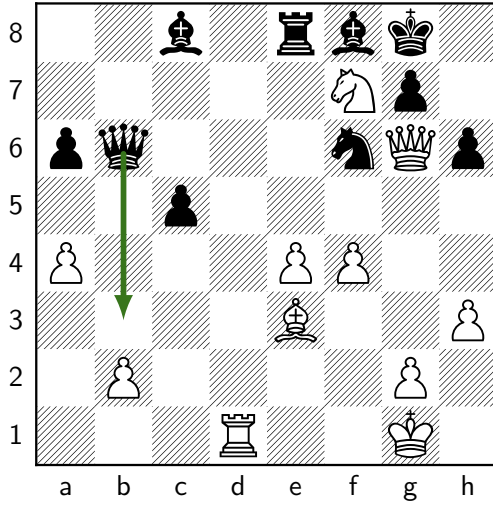
Figure 15: **Move 21, White: Qc1.** (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#885 (ply 40) [structural distance $d=94$, QAP obj=64]: the master played Rg6; under μ our Qd1 plays the role of that game’s Rg1 (a **cross-type** correspondence); cross-type roles: our nc6 as their Rd1, our pc7 as their Bf1; transferred move d1c1 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#428 (ply 57) [structural distance $d=89$, QAP obj=50]: the master played Qe8; under μ our Qd1 plays the role of that game’s Qg3; cross-type roles: our Pd4 as their Rb8, our Bg2 as their Pg7; transferred move d1c1 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2328 (ply 39) [structural distance $d=95$, QAP obj=38]: the master played Rb8; under μ our Qd1 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ba8, our Pb2 as their Be7; transferred move d1c1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qc1 (summed QAP objective 152, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



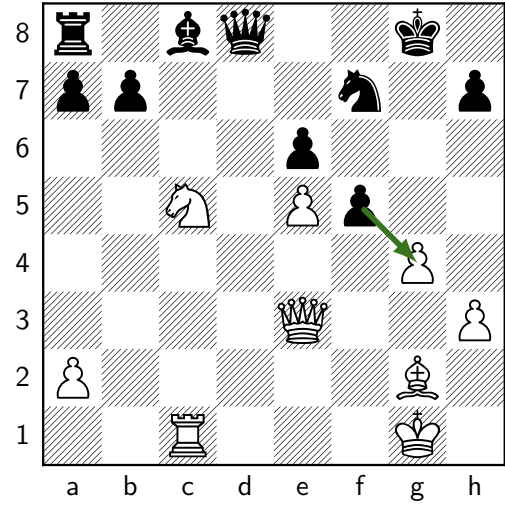
(a)



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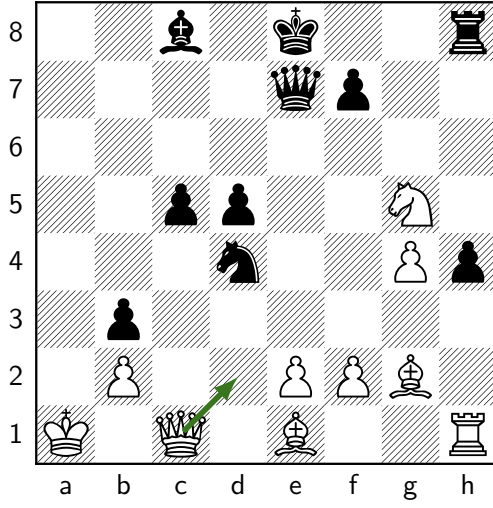


(c)

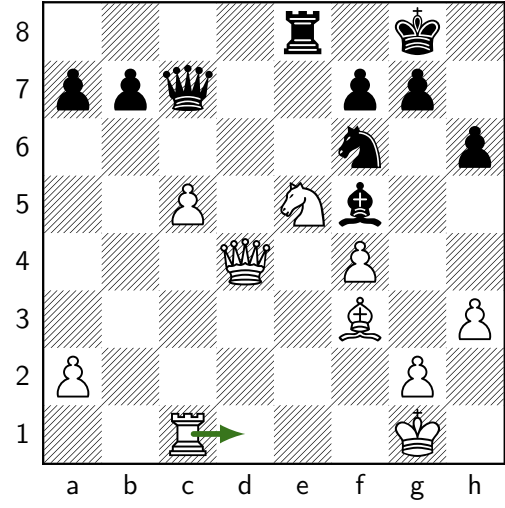


(d)

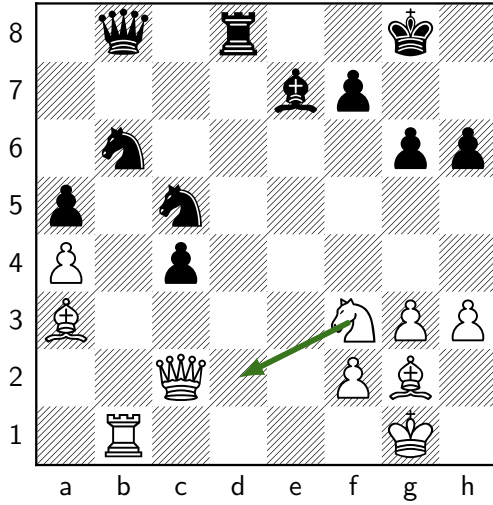
Figure 16: **Move 22, White: h3g4.** (a) Our position; Melanie plays h3g4 (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#740` (ply 48) [structural distance $d=70$, QAP obj=80]: the master played Qxd4; under μ our $Ph3$ plays the role of that game's $Qb2$ (a **cross-type** correspondence); cross-type roles: our $Qc1$ as their $Rc1$, our $Be1$ as their $Pa2$; transferred move h3g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3913` (ply 49) [structural distance $d=56$, QAP obj=36]: the master played Qb3; under μ our $Ph3$ plays the role of that game's $Qb3$ (a **cross-type** correspondence); cross-type roles: our $Qc1$ as their $Bc8$, our $Bg2$ as their $Re8$; transferred move h3g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#451` (ply 57) [structural distance $d=85$, QAP obj=35]: the master played fxg4; under μ our $Ph3$ plays the role of that game's $Pf4$; cross-type roles: our $Pb2$ as their $Ra8$, our $Ng5$ as their $Bc8$; transferred move h3g4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to h3g4 (summed QAP objective 195, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



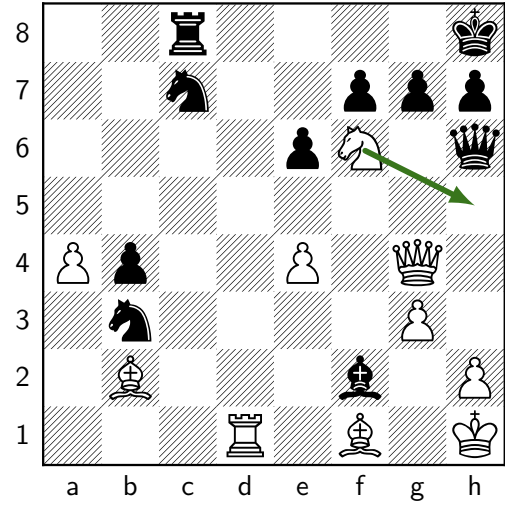
(a)



(b)

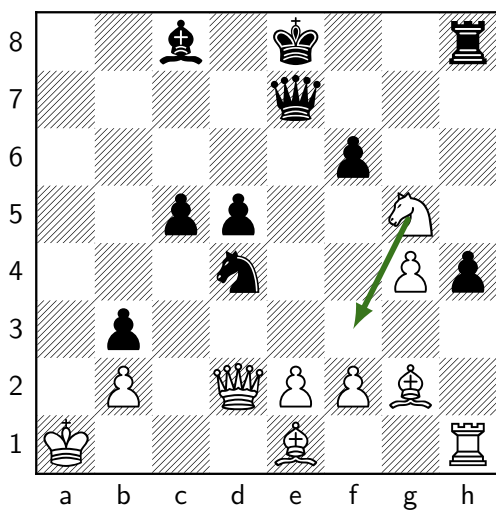


(c)

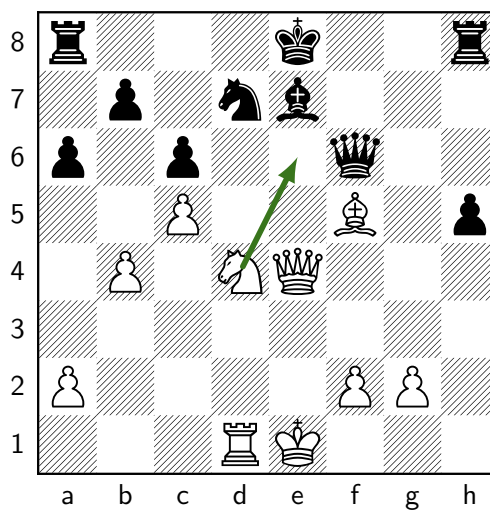


(d)

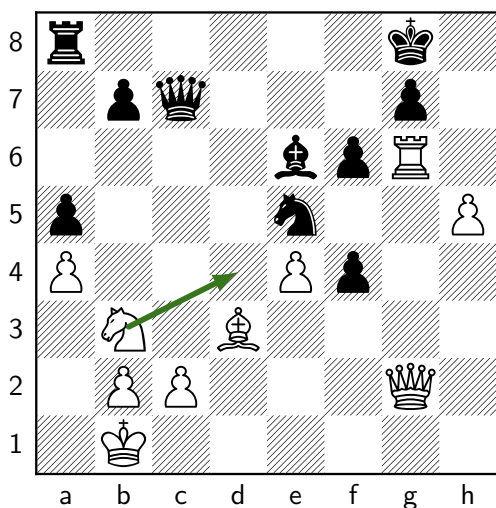
Figure 17: **Move 23, White: Qd2.** (a) Our position; Melanie plays Qd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#740` (ply 54) [structural distance $d=82$, QAP obj=87]: the master played Rd1; under μ our Qc1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Be1 as their Pa2, our Rh1 as their Bf3; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#862` (ply 60) [structural distance $d=83$, QAP obj=68]: the master played Nd2; under μ our Qc1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our qe7 as their Rb1, our Rh1 as their Qc2; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2853` (ply 58) [structural distance $d=65$, QAP obj=76]: the master played Nh5; under μ our Ng5 plays the role of that game’s Nf6; cross-type roles: our Bg2 as their Rd1, our Pg4 as their Bf1; transferred move g5e4. Synthesis: 2 retrieved analogs transferred to Qd2 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



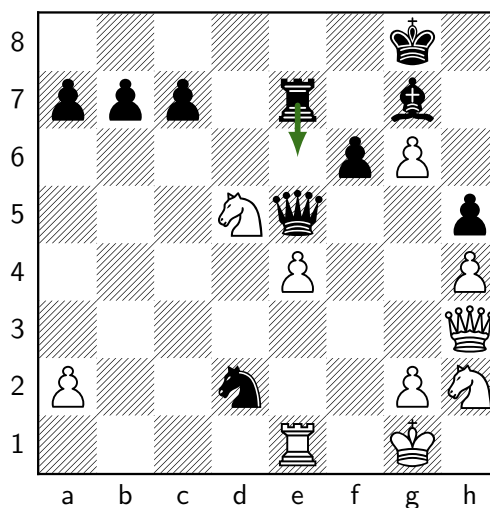
(a)



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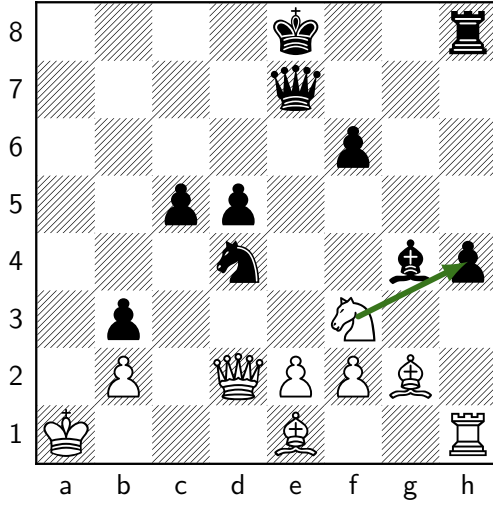


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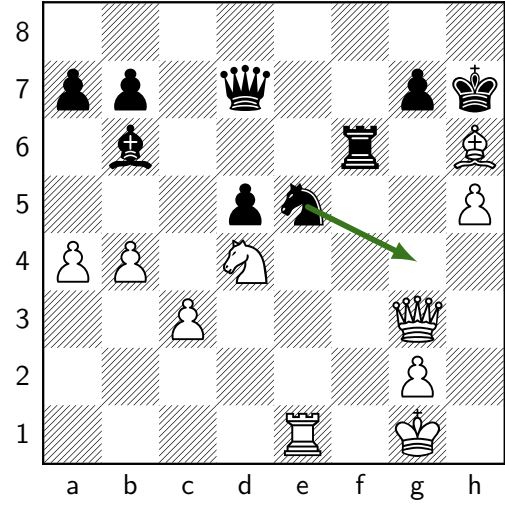


(d)

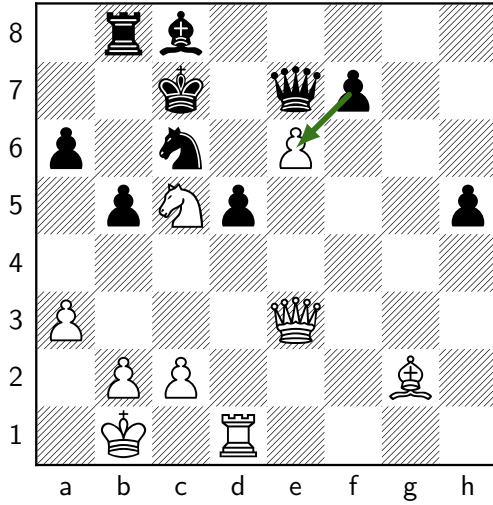
Figure 18: **Move 24, White: Nf3**. (a) Our position; Melanie plays Nf3 (g5f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1572` (ply 50) [structural distance $d=85$, QAP obj=45]: the master played Ne6; under μ our Ng5 plays the role of that game’s Nd4; cross-type roles: our Pb2 as their Rd1, our Be1 as their Pa2; transferred move g5f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#381` (ply 52) [structural distance $d=87$, QAP obj=45]: the master played Nd4; under μ our Ng5 plays the role of that game’s Nb3; cross-type roles: our pc5 as their Qg2, our nd4 as their Bd3; transferred move g5f3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2459` (ply 59) [structural distance $d=74$, QAP obj=36]: the master played Re6; under μ our Ng5 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our Be1 as their Pa7, our Rh1 as their Pb7; transferred move g5f3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nf3 (summed QAP objective 187, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



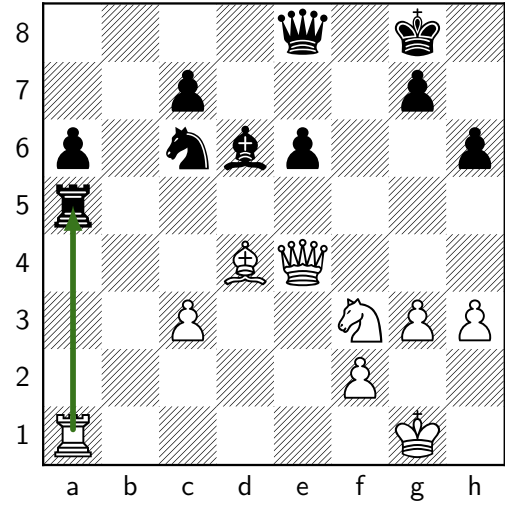
(a)



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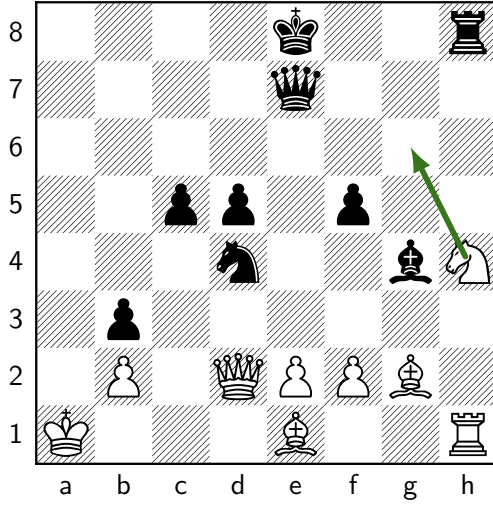


(c)

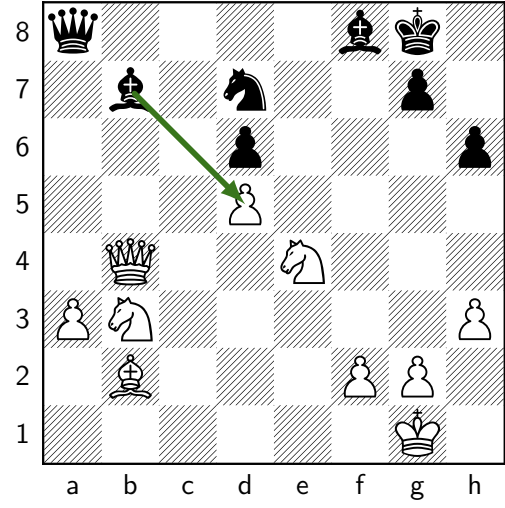


(d)

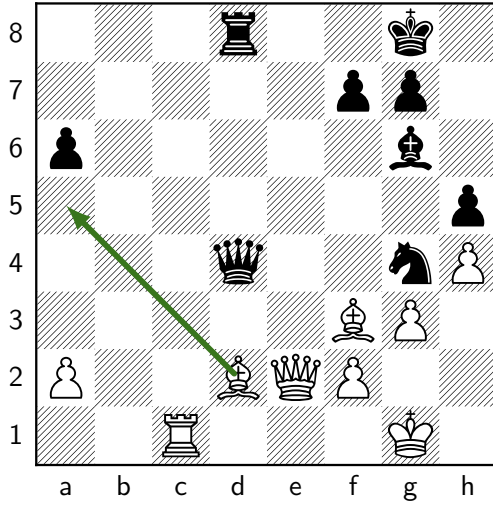
Figure 19: **Move 25, White: Nxh4.** (a) Our position; Melanie plays Nxh4 (f3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#428 (ply 57) [structural distance $d=97$, QAP obj=52]: the master played Ng4; under μ our Nf3 plays the role of that game’s Ne4; cross-type roles: our Be1 as their Pa7, our Bg2 as their Qd7; transferred move f3h4 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2971 (ply 49) [structural distance $d=102$, QAP obj=43]: the master played fxe6; under μ our Nf3 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Bc8, our Be1 as their Pa6; transferred move f3h4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3956 (ply 52) [structural distance $d=89$, QAP obj=38]: the master played Rxa5; under μ our Nf3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Bg2 as their Nf3, our Qd2 as their Pg3; transferred move f3h4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nxh4 (summed QAP objective 133, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



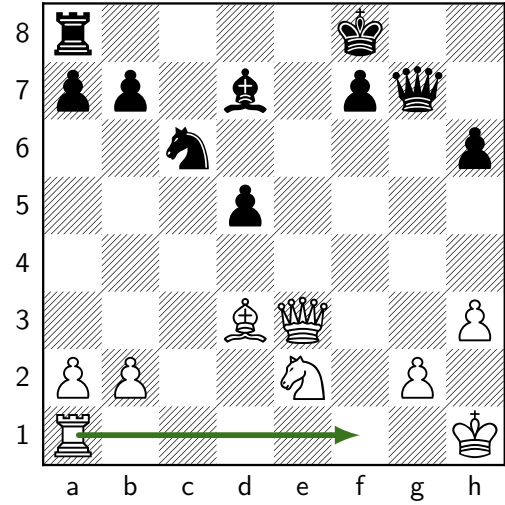
(a)



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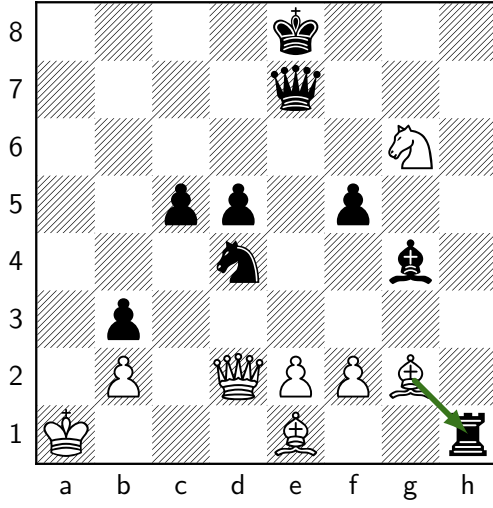


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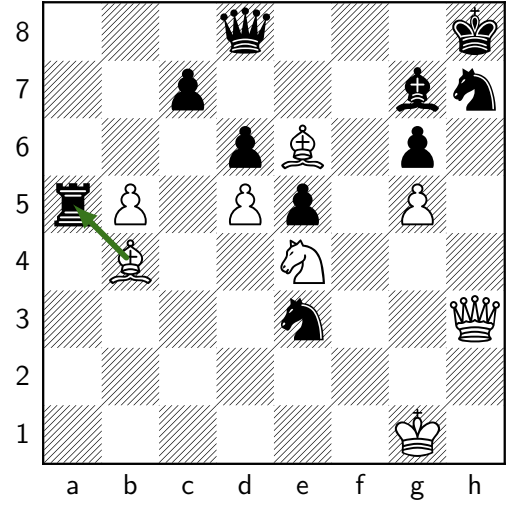


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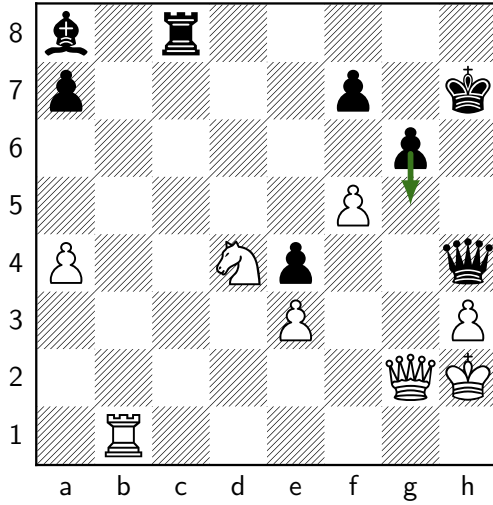
Figure 20: **Move 26, White: Ng6**. (a) Our position; Melanie plays Ng6 (h4g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1670` (ply 57) [structural distance $d=81$, QAP obj=56]: the master played Bxd5; under μ our Nh4 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Nd7, our bg4 as their nb3; transferred move h4g6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2454` (ply 50) [structural distance $d=82$, QAP obj=54]: the master played Ba5; under μ our Nh4 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our Be1 as their Pa2, our Pb2 as their Qe2; transferred move h4g6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1278` (ply 44) [structural distance $d=81$, QAP obj=19]: the master played Rf1; under μ our Nh4 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pa2, our Bg2 as their Ne2; transferred move h4g6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ng6 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



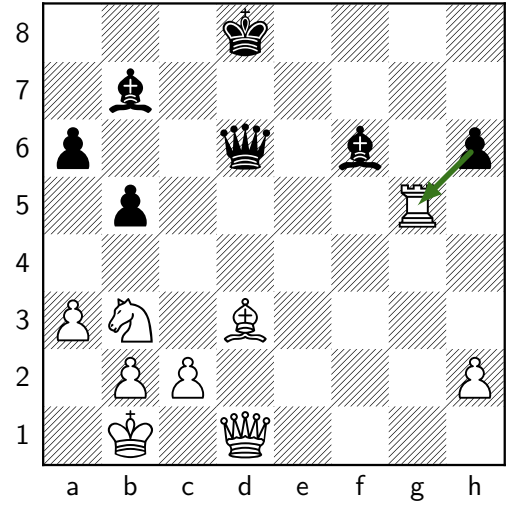
(a)



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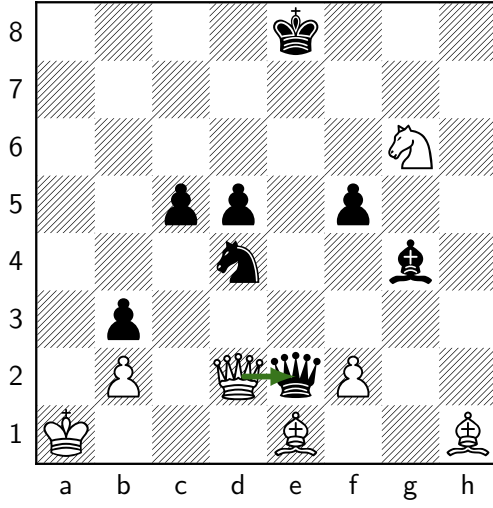


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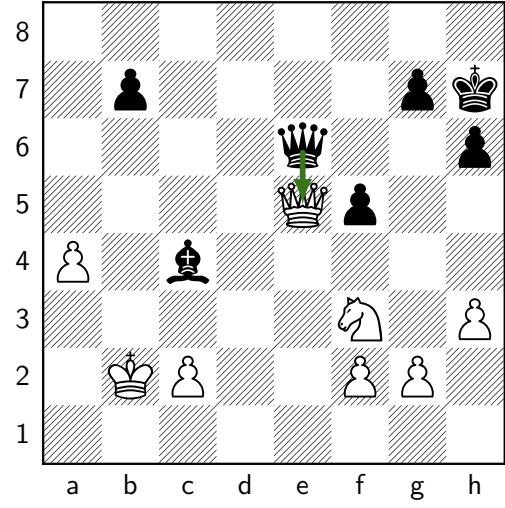


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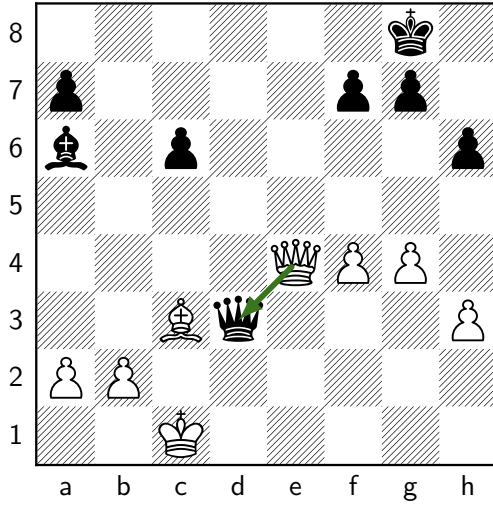
Figure 21: **Move 27, White: Bxh1.** (a) Our position; Melanie plays Bxh1 (g2h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#849` (ply 64) [structural distance $d=77$, QAP obj=40]: the master played Bxa5; under μ our Bg2 plays the role of that game’s Bb4; cross-type roles: our pb3 as their ne3, our pc5 as their Ne4; transferred move g2h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1904` (ply 67) [structural distance $d=78$, QAP obj=28]: the master played g5; under μ our Bg2 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our Qd2 as their Ba8, our Pb2 as their Rc8; transferred move g2h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#379` (ply 61) [structural distance $d=73$, QAP obj=52]: the master played hxg5; under μ our Ng6 plays the role of that game’s Ph3 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Bb7, our rh1 as their Qd6; transferred move g6e5. Synthesis: 2 retrieved analogs transferred to Bxh1 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



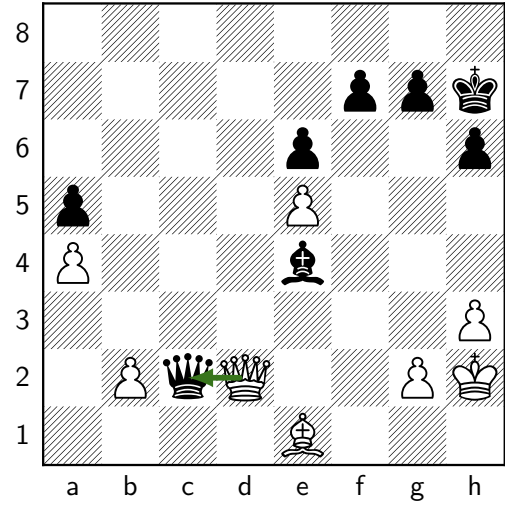
(a)



(b)



(c)



(d)

Figure 22: **Move 28, White: Qxe2+.** (a) Our position; Melanie plays Qxe2+ (d2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3000` (ply 57) [structural distance $d=76$, QAP obj=59]: the master played Qxe5+; under μ our Qd2 plays the role of that game's Qe3; cross-type roles: our Be1 as their Pf5, our Ng6 as their Bc4; transferred move d2e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3219` (ply 50) [structural distance $d=78$, QAP obj=56]: the master played Qxd3; under μ our Qd2 plays the role of that game's Qe4; cross-type roles: our nd4 as their pc6, our Bh1 as their ph6; transferred move d2e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2706` (ply 68) [structural distance $d=82$, QAP obj=51]: the master played Qxc2; under μ our Qd2 plays the role of that game's Qd2; cross-type roles: our Bh1 as their Ph3, our nd4 as their pa5; transferred move d2e2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qxe2+ (summed QAP objective 194, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

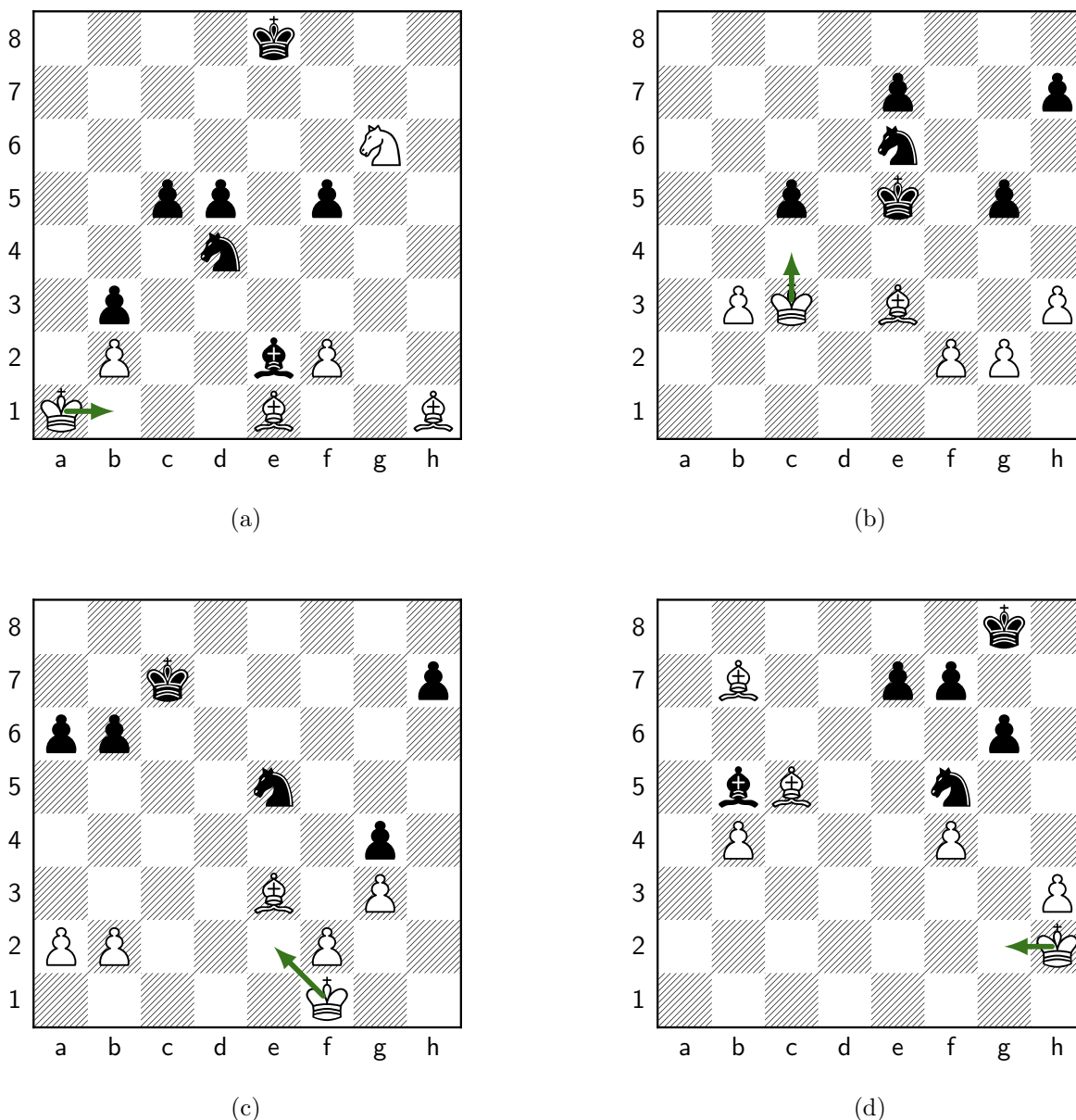


Figure 23: **Move 29, White: Kb1.** (a) Our position; Melanie plays Kb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#987` (ply 68) [structural distance $d=28$, QAP obj=16]: the master played Kc4; under μ our Ka1 plays the role of that game’s Kc3; cross-type roles: our Be1 as their Pg2, our be2 as their ph7; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3093` (ply 72) [structural distance $d=37$, QAP obj=16]: the master played Ke2; under μ our Ka1 plays the role of that game’s Kf1; cross-type roles: our Be1 as their Pb2, our be2 as their pa6; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1028` (ply 72) [structural distance $d=24$, QAP obj=13]: the master played Kg2; under μ our Ka1 plays the role of that game’s Kh2; cross-type roles: our Be1 as their Pf4, our Pf2 as their bb5; transferred move a1b1 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Kb1 (summed QAP objective 173, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

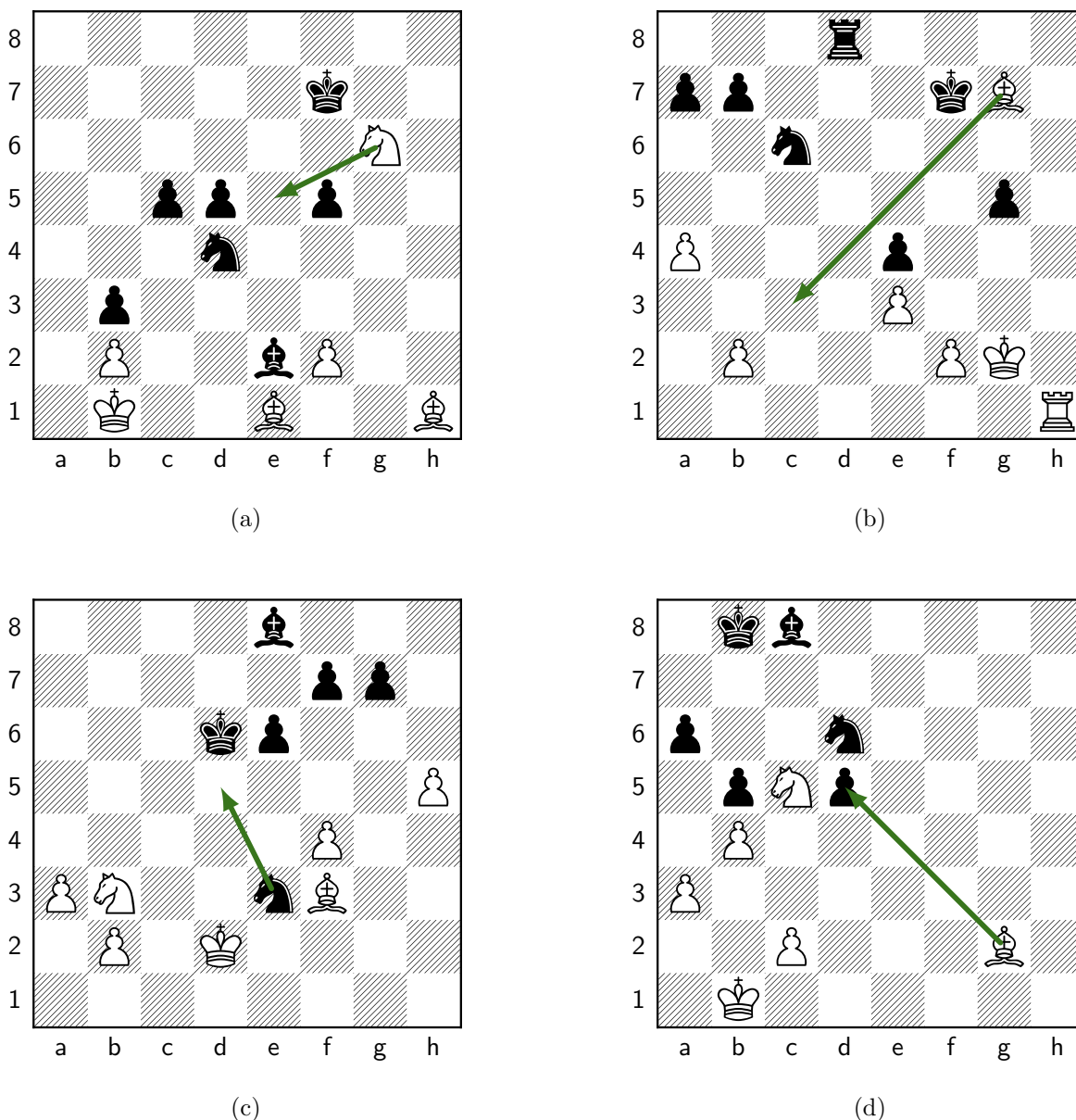
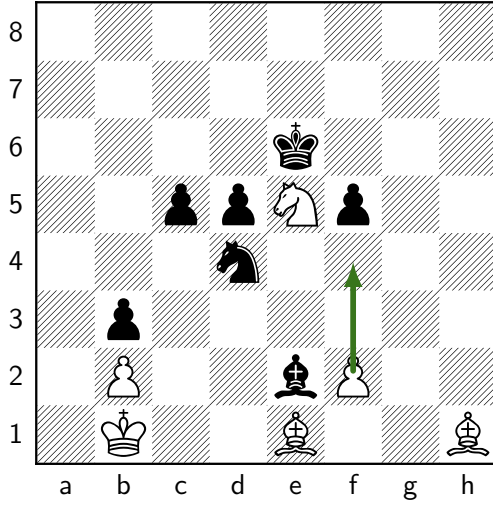
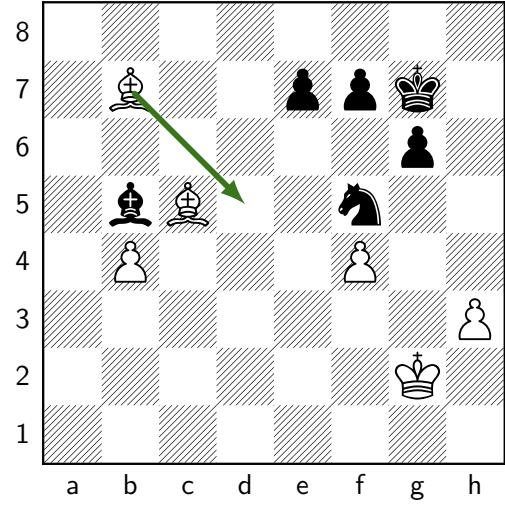


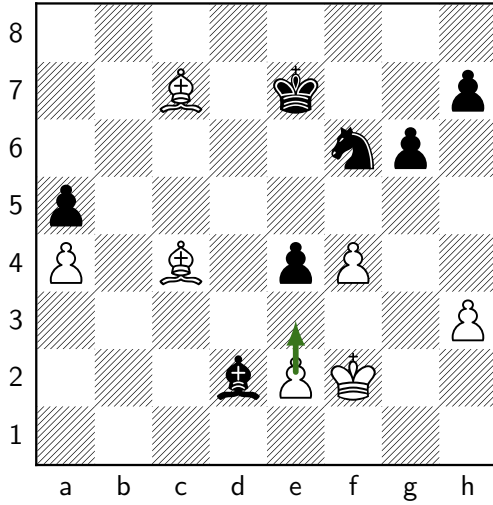
Figure 24: **Move 30, White: Ne5+ (g6e5).** (a) Our position; Melanie plays Ne5+ (g6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2462` (ply 62) [structural distance $d=39$, QAP obj=20]: the master played Bc3; under μ our Ng6 plays the role of that game’s Bg7 (a **cross-type** correspondence); cross-type roles: our Be1 as their Rh1, our Bh1 as their Pb2; transferred move g6e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2814` (ply 73) [structural distance $d=29$, QAP obj=19]: the master played Nd5; under μ our Ng6 plays the role of that game’s Ne6; cross-type roles: our pc5 as their Be8, our Bh1 as their Pf7; transferred move g6e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2971` (ply 74) [structural distance $d=38$, QAP obj=3]: the master played Bxd5; under μ our Ng6 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our nd4 as their Pb4, our be2 as their Nc5; transferred move g6e5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ne5+ (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



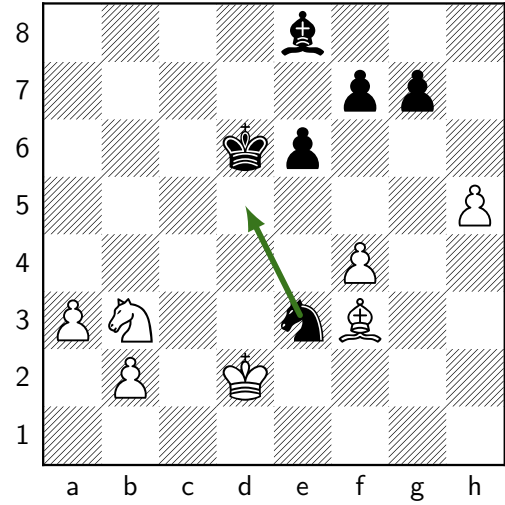
(a)



(b)

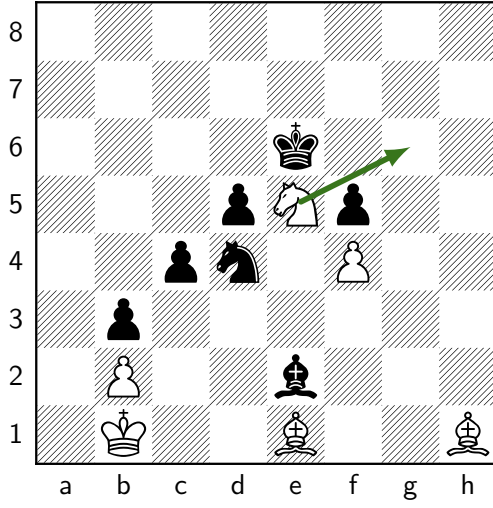


(c)

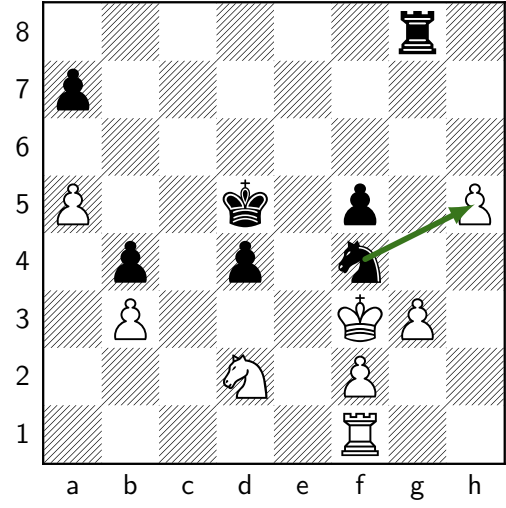


(d)

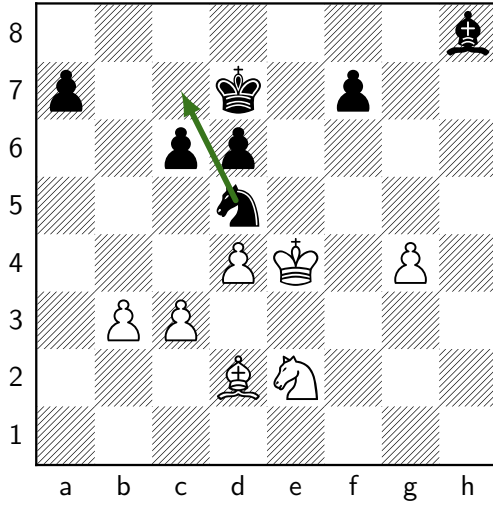
Figure 25: **Move 31, White: f4.** (a) Our position; Melanie plays f4 (f2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1028` (ply 74) [structural distance $d=41$, QAP obj=4]: the master played Bd5; under μ our Pf2 plays the role of that game’s Bb7 (a **cross-type** correspondence); cross-type roles: our Ne5 as their Bc5, our Bh1 as their pg6; transferred move f2f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2008` (ply 66) [structural distance $d=44$, QAP obj=1]: the master played e3; under μ our Pf2 plays the role of that game’s Pe2; cross-type roles: our Bh1 as their Pa4, our nd4 as their Bc4; transferred move f2f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2814` (ply 73) [structural distance $d=51$, QAP obj=22]: the master played Nd5; under μ our Ne5 plays the role of that game’s Ne6; cross-type roles: our Bh1 as their Pg7; transferred move e5c6. Synthesis: 2 retrieved analogs transferred to f4 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



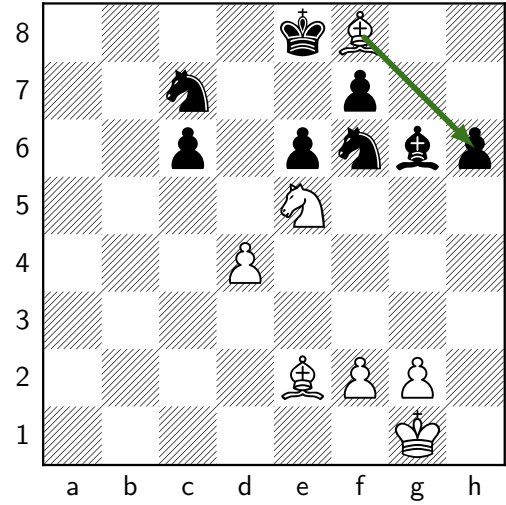
(a)



(b)

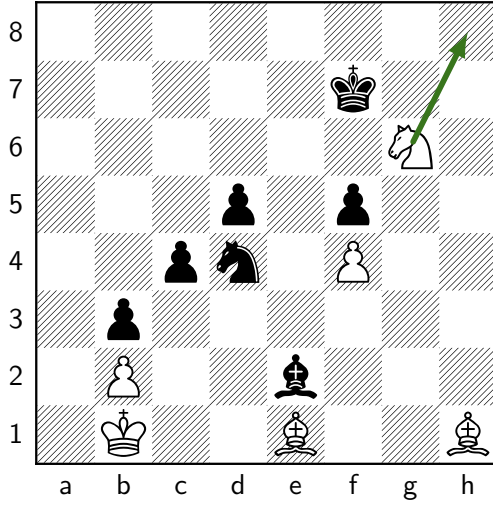


(c)

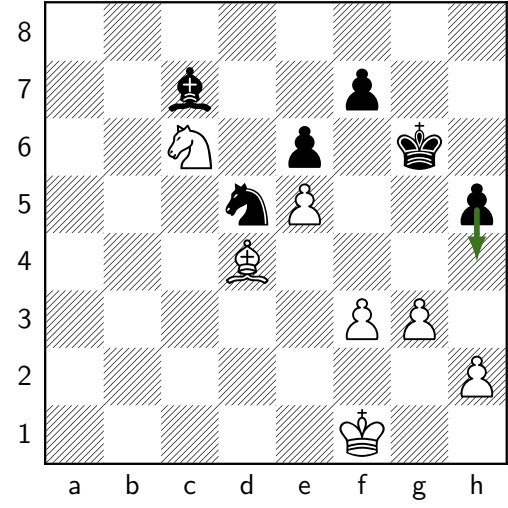


(d)

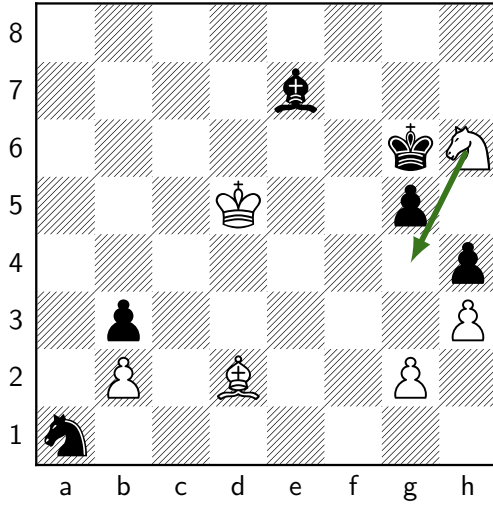
Figure 26: **Move 32, White: Ng6.** (a) Our position; Melanie plays Ng6 (e5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#568` (ply 71) [structural distance $d=61$, QAP obj=58]: the master played Nxb5; under μ our Ne5 plays the role of that game’s Nf5; cross-type roles: our Bh1 as their Rg8, our Be1 as their Pb4; transferred move e5g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1760` (ply 63) [structural distance $d=43$, QAP obj=39]: the master played Nc7; under μ our Ne5 plays the role of that game’s Nd4; cross-type roles: our Be1 as their Pa7; transferred move e5g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#715` (ply 58) [structural distance $d=51$, QAP obj=33]: the master played Bxb6; under μ our Ne5 plays the role of that game’s Bf8 (a **cross-type** correspondence); cross-type roles: our pb3 as their Be2, our Be1 as their Pf2; transferred move e5g6 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Ng6 (summed QAP objective 258, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



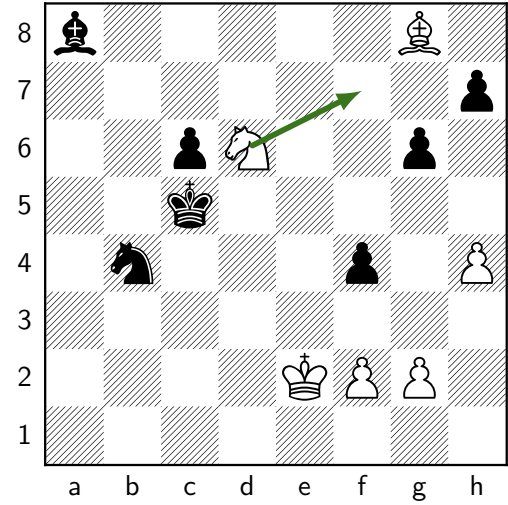
(a)



(b)

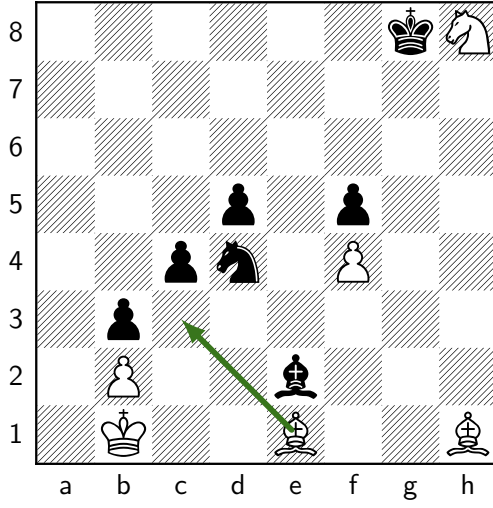


(c)

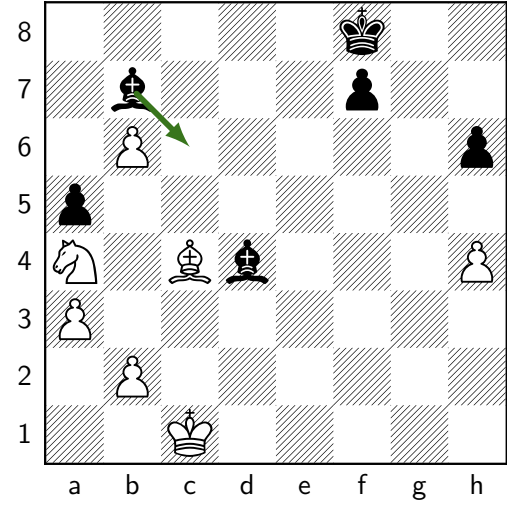


(d)

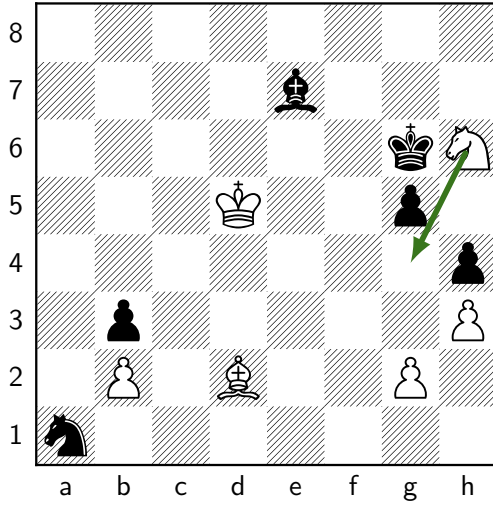
Figure 27: **Move 33, White: Nh8+ (g6h8).** (a) Our position; Melanie plays Nh8+ (g6h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#277` (ply 77) [structural distance $d=39$, QAP obj=2]: the master played **h4**; under μ our Ng6 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our pd5 as their Bc7, our Pf4 as their Nd5; transferred move **g6h8 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#3769` (ply 72) [structural distance $d=38$, QAP obj=21]: the master played Ng4; under μ our Ng6 plays the role of that game’s Nh6; cross-type roles: our be2 as their pb3, our pf5 as their be7; transferred move **g6h4**. (d) Retrieved from `ccr1_top_p0003#2584` (ply 70) [structural distance $d=31$, QAP obj=20]: the master played Nf7; under μ our Ng6 plays the role of that game’s Nd6; cross-type roles: our Be1 as their Pg2, our pd5 as their nb4; transferred move **g6h4**. Synthesis: 1 retrieved analog transferred to Nh8+ (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



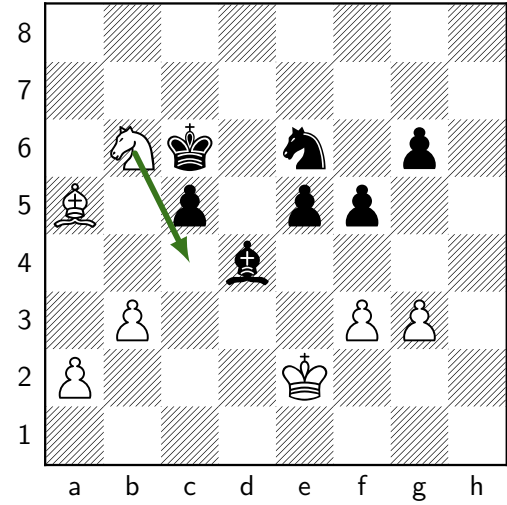
(a)



(b)

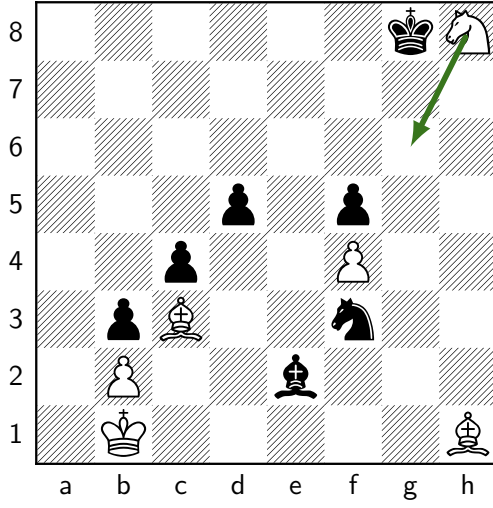


(c)

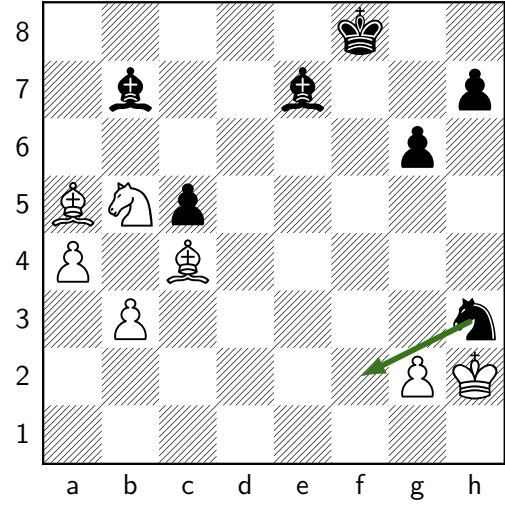


(d)

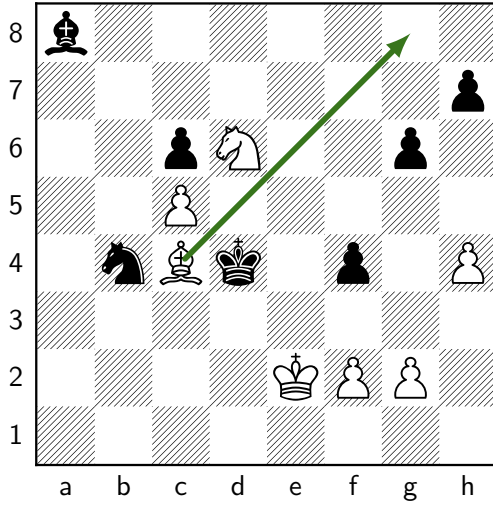
Figure 28: **Move 34, White: Bc3.** (a) Our position; Melanie plays Bc3 (e1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#717` (ply 65) [structural distance $d=27$, QAP obj=3]: the master played Bc6; under μ our Be1 plays the role of that game’s Bb2; cross-type roles: our Nh8 as their Pa5, our pb3 as their bc4; transferred move e1c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3769` (ply 72) [structural distance $d=31$, QAP obj=21]: the master played Ng4; under μ our Nh8 plays the role of that game’s Nh6; cross-type roles: our be2 as their pb3, our pf5 as their be7; transferred move h8g6. (d) Retrieved from `ccr1_top_p0003#2510` (ply 82) [structural distance $d=37$, QAP obj=21]: the master played Nc4; under μ our Nh8 plays the role of that game’s Nb6; cross-type roles: our Be1 as their Pa2; transferred move h8g6. Synthesis: 1 retrieved analog transferred to Bc3 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



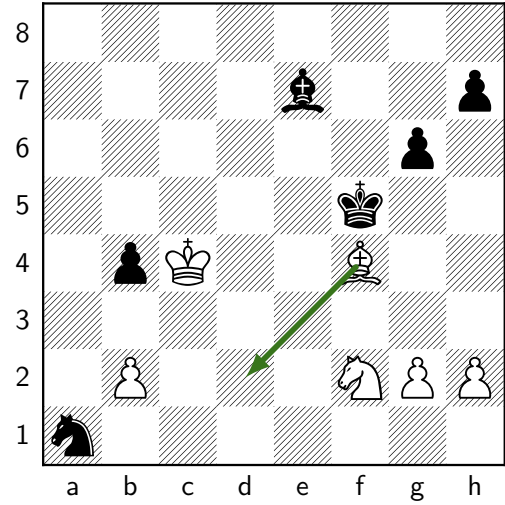
(a)



(b)

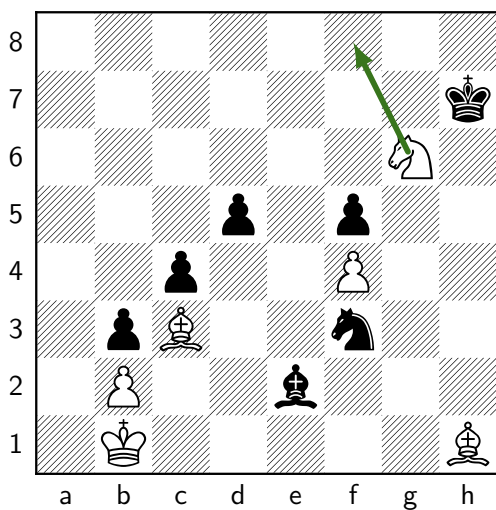


(c)

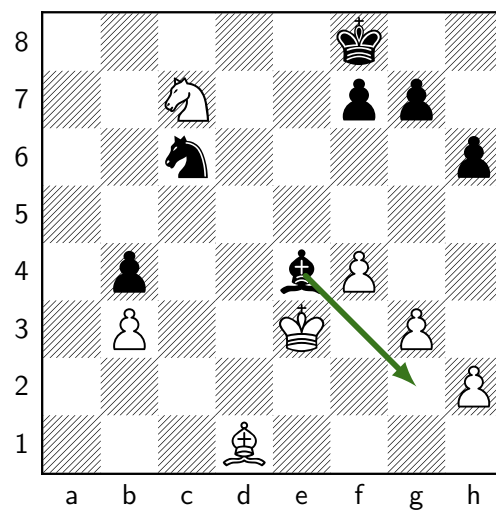


(d)

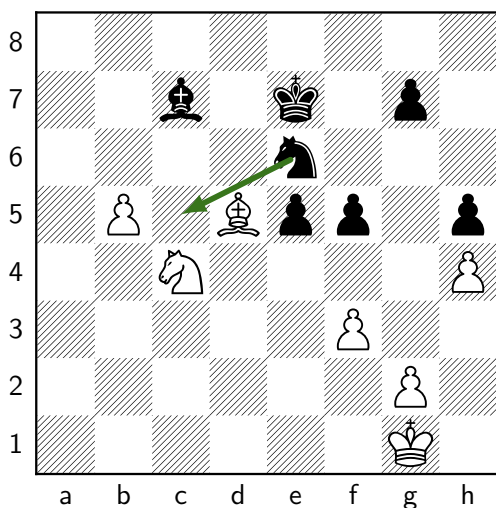
Figure 29: **Move 35, White: Ng6.** (a) Our position; Melanie plays Ng6 (h8g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2397` (ply 67) [structural distance $d=31$, QAP obj=33]: the master played Nf2; under μ our Nh8 plays the role of that game’s Nh6; cross-type roles: our pb3 as their Be7, our Bh1 as their Ph7; transferred move h8g6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2584` (ply 68) [structural distance $d=35$, QAP obj=22]: the master played Bg8; under μ our Nh8 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our be2 as their Pf2, our pd5 as their nb4; transferred move h8g6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3769` (ply 62) [structural distance $d=34$, QAP obj=19]: the master played Bd2; under μ our Nh8 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our Bc3 as their na1, our pd5 as their Nf2; transferred move h8g6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Ng6 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



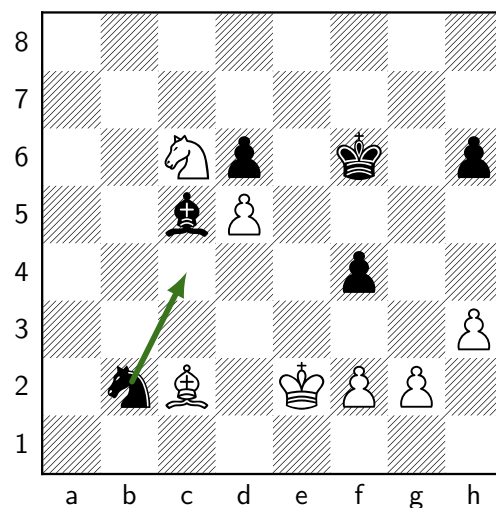
(a)



(b)

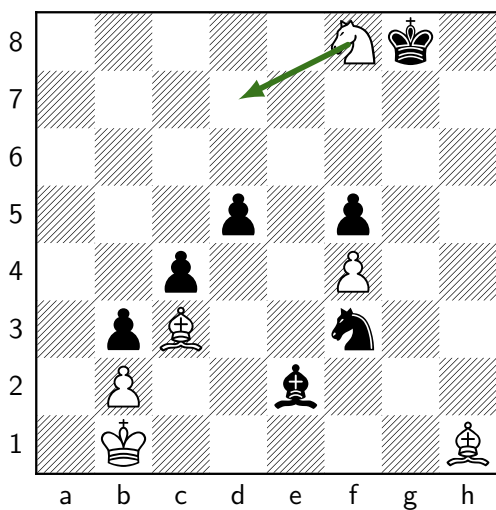


(c)

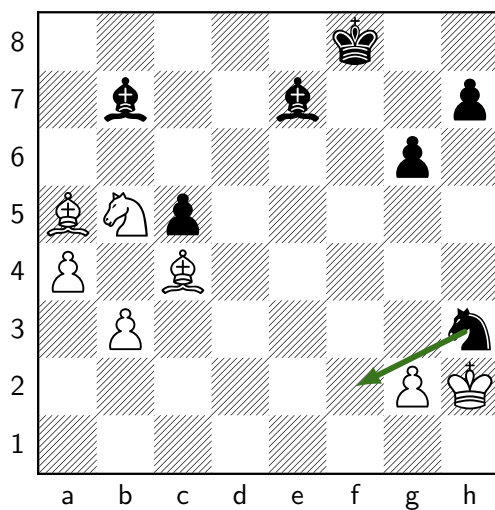


(d)

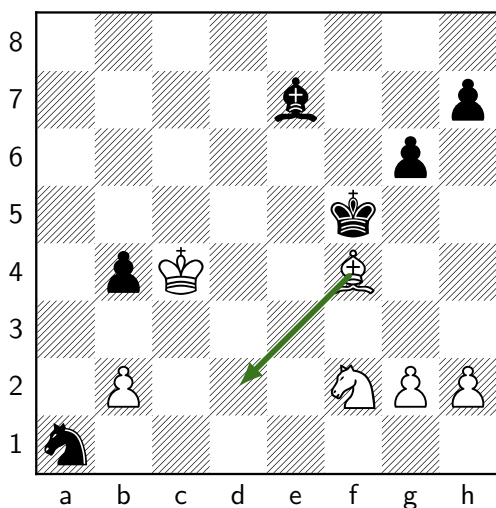
Figure 30: **Move 36, White: Nf8+ (g6f8).** (a) Our position; Melanie plays Nf8+ (g6f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3029` (ply 85) [structural distance $d=32$, QAP obj=21]: the master played Bg2; under μ our Ng6 plays the role of that game’s Be5 (a **cross-type** correspondence); cross-type roles: our Bc3 as their nc7, our Bh1 as their Pg7; transferred move g6f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3303` (ply 69) [structural distance $d=30$, QAP obj=15]: the master played Nc5; under μ our Ng6 plays the role of that game’s Ne3; cross-type roles: our Bc3 as their Pg7, our kh7 as their bd5; transferred move g6f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3798` (ply 87) [structural distance $d=36$, QAP obj=4]: the master played Nc4; under μ our Ng6 plays the role of that game’s Nb7; cross-type roles: our pb3 as their nc6, our Bh1 as their Ph6; transferred move g6f8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nf8+ (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



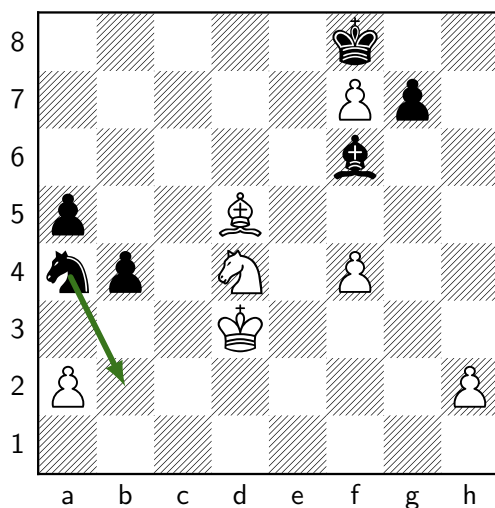
(a)



(b)

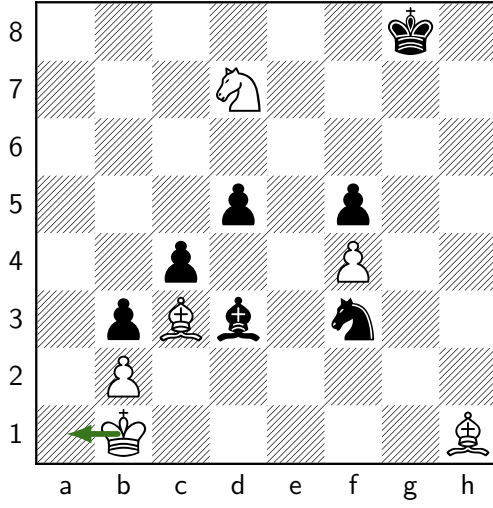


(c)

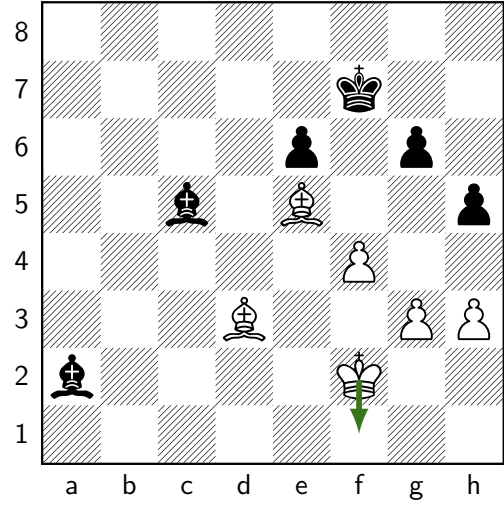


(d)

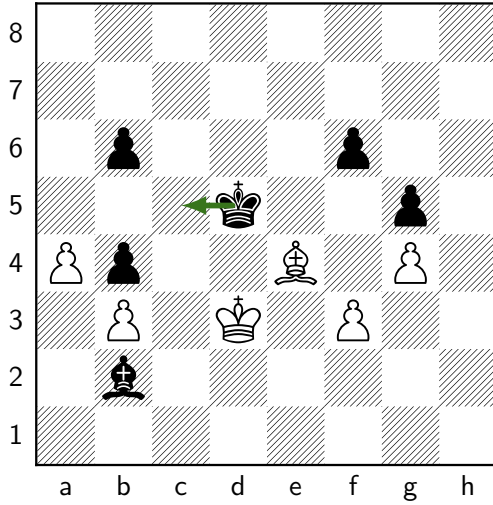
Figure 31: **Move 37, White: Nd7.** (a) Our position; Melanie plays Nd7 (f8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2397` (ply 67) [structural distance $d=31$, QAP obj=33]: the master played Nf2; under μ our Nf8 plays the role of that game’s Nh6; cross-type roles: our Pf4 as their Be7, our Bh1 as their Ph7; transferred move f8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3769` (ply 62) [structural distance $d=30$, QAP obj=19]: the master played Bd2; under μ our Nf8 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our pf5 as their na1, our Bc3 as their Nf2; transferred move f8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1189` (ply 65) [structural distance $d=26$, QAP obj=11]: the master played Nb2+; under μ our Nf8 plays the role of that game’s Na5; cross-type roles: our Bc3 as their pf4; transferred move f8d7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nd7 (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



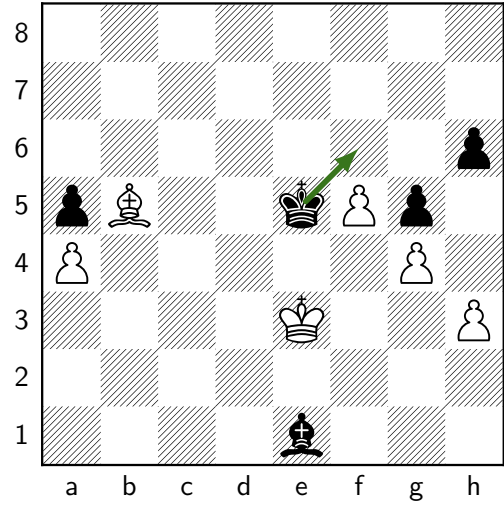
(a)



(b)



(c)



(d)

Figure 32: **Move 38, White: Ka1.** (a) Our position; Melanie plays **Ka1** (**b1a1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#2735` (ply 82) [structural distance $d=28$, QAP obj=50]: the master played **Kf1**; under μ our **Kb1** plays the role of that game’s **Kf2**; cross-type roles: our **pf5** as their **Bd3**, our **Bc3** as their **Pf4**; transferred move **b1a1** (**endorses this move**). (c) Retrieved from `cctl_top_p0001#383` (ply 85) [structural distance $d=31$, QAP obj=35]: the master played **Kc5**; under μ our **Kb1** plays the role of that game’s **Kd4**; cross-type roles: our **nf3** as their **Pb6**, our **Bc3** as their **pg4**; transferred move **b1a1** (**endorses this move**). (d) Retrieved from `cctl_top_p0000#292` (ply 87) [structural distance $d=30$, QAP obj=4]: the master played **Kf6**; under μ our **Kb1** plays the role of that game’s **Ke4**; cross-type roles: our **bd3** as their **pf5**, our **nf3** as their **Pg5**; transferred move **b1a1** (**endorses this move**). Synthesis: 12 retrieved analogs transferred to **Ka1** (summed QAP objective 111, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

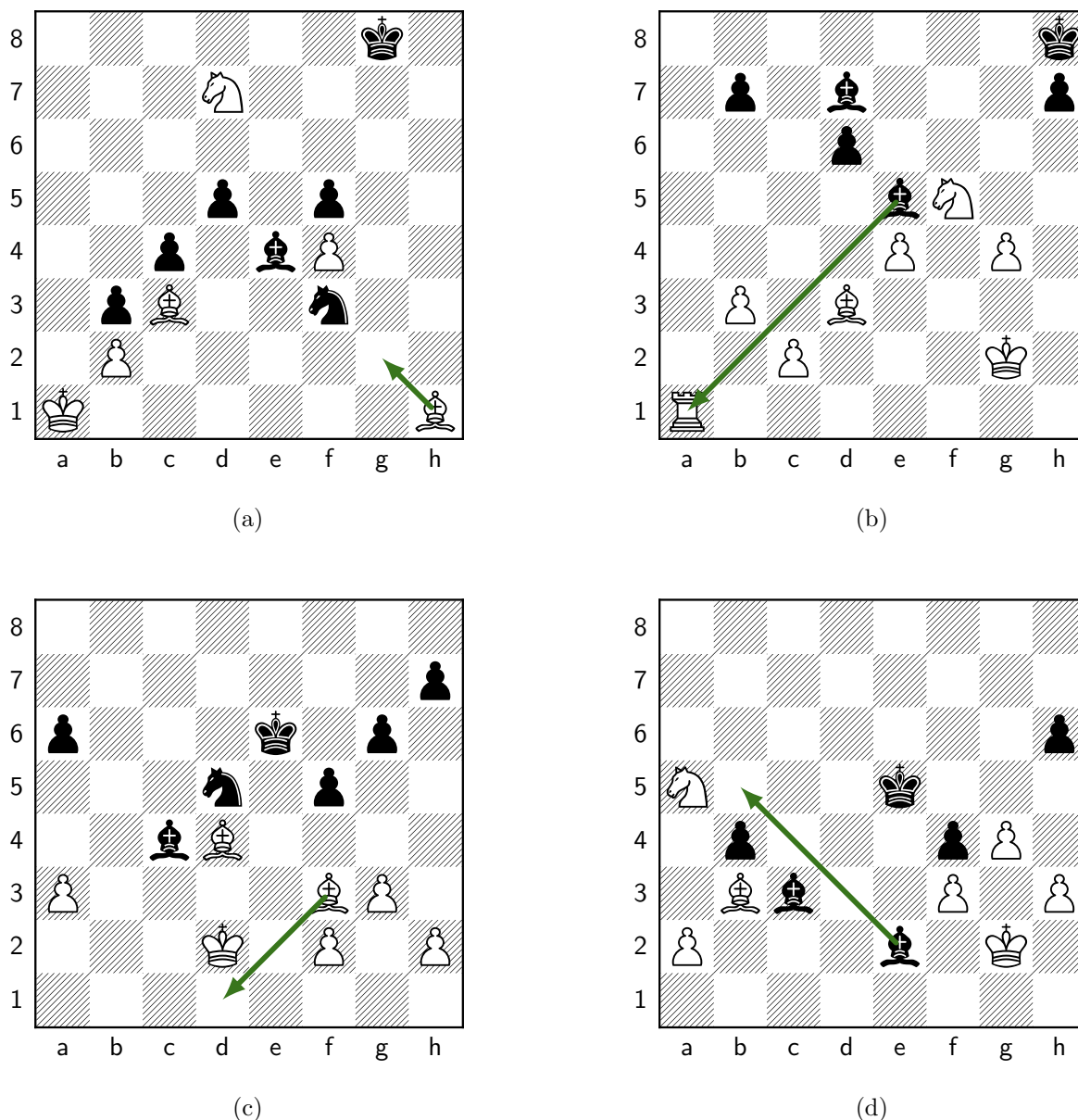
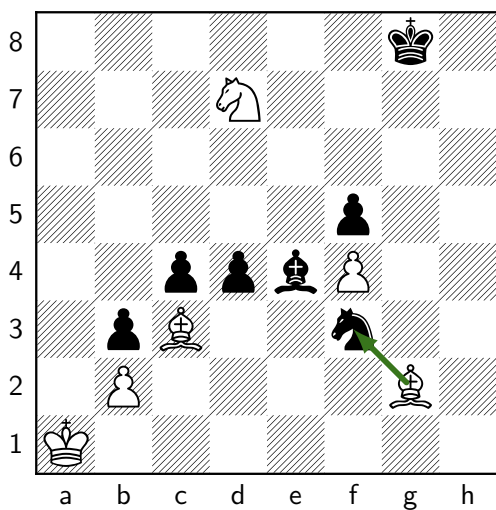
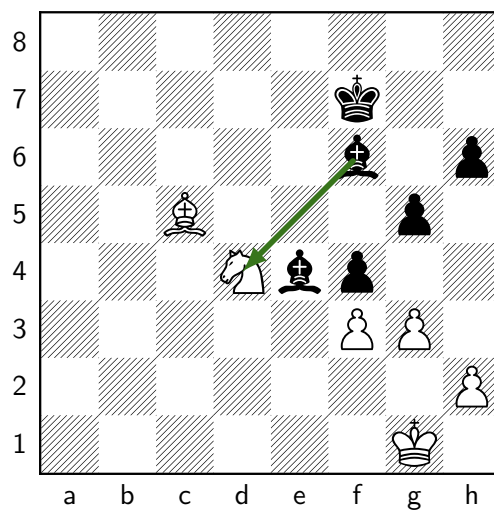


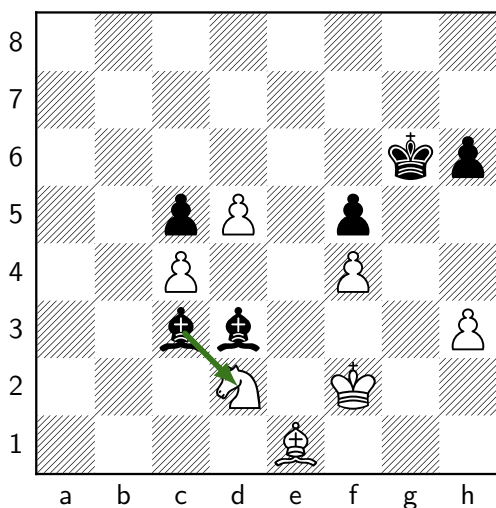
Figure 33: **Move 39, White: Bg2.** (a) Our position; Melanie plays Bg2 (h1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2615 (ply 67) [structural distance $d=27$, QAP obj=26]: the master played Bxa1; under μ our Bh1 plays the role of that game's Be4; cross-type roles: our Bc3 as their Ph7, our be4 as their pe4; transferred move h1g2 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3700 (ply 62) [structural distance $d=38$, QAP obj=24]: the master played Bd1; under μ our Bh1 plays the role of that game's Bf3; cross-type roles: our Bc3 as their Pg3, our Nd7 as their Bd4; transferred move h1g2 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1964 (ply 87) [structural distance $d=33$, QAP obj=15]: the master played Bb5; under μ our Bh1 plays the role of that game's Be7; cross-type roles: our Pb2 as their na5, our Bc3 as their Pf4; transferred move h1g2 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Bg2 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



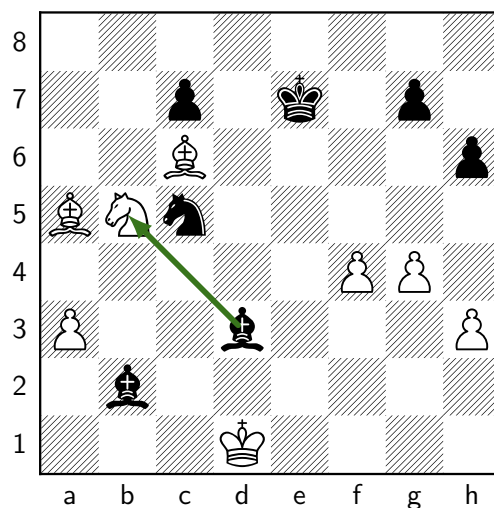
(a)



(b)

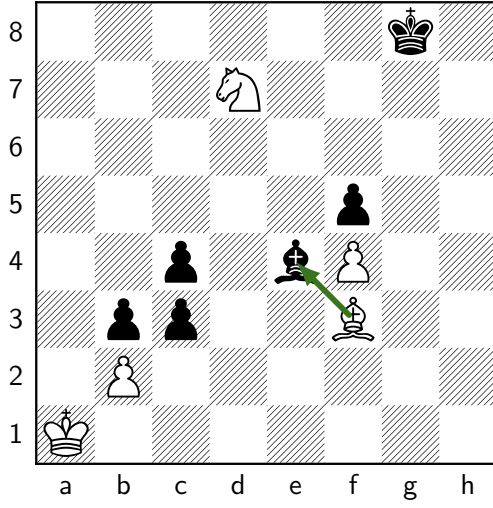


(c)

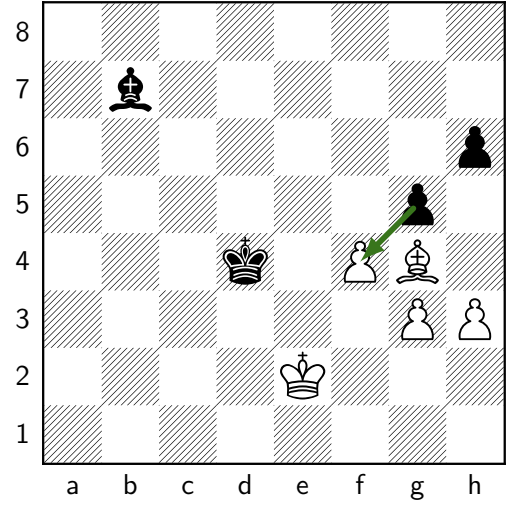


(d)

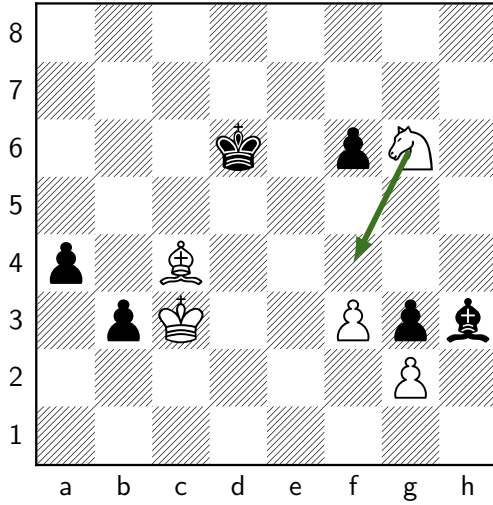
Figure 34: **Move 40, White: Bxf3**. (a) Our position; Melanie plays Bxf3 (g2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1252` (ply 79) [structural distance $d=28$, QAP obj=46]: the master played Bxd4+; under μ our Bg2 plays the role of that game’s Bf3; cross-type roles: our Bc3 as their Pf4; transferred move g2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1220` (ply 89) [structural distance $d=23$, QAP obj=30]: the master played Bxd2; under μ our Bg2 plays the role of that game’s Bc6; cross-type roles: our Nd7 as their pd5; transferred move g2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2775` (ply 69) [structural distance $d=31$, QAP obj=23]: the master played Bxb5; under μ our Bg2 plays the role of that game’s Bd6; cross-type roles: our pf5 as their ba5; transferred move g2f3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bxf3 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



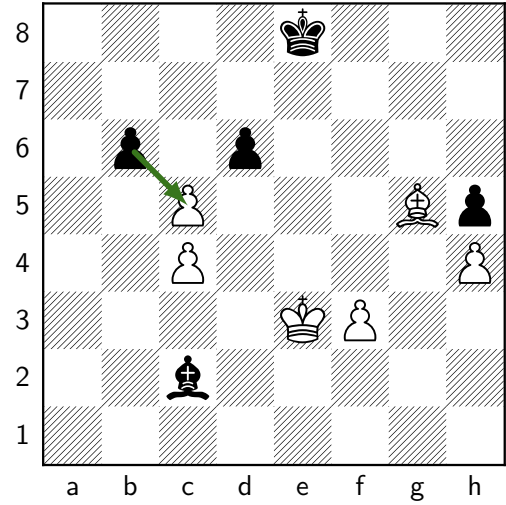
(a)



(b)

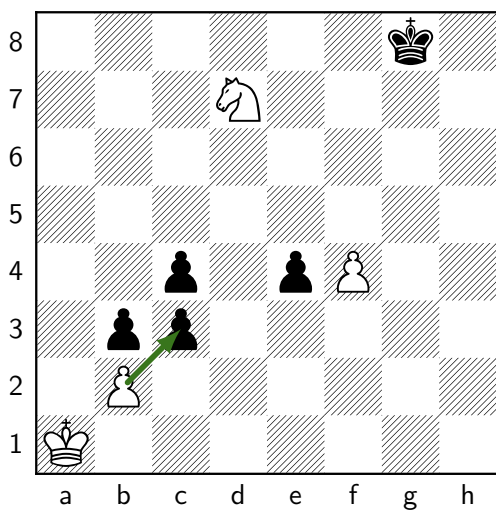


(c)

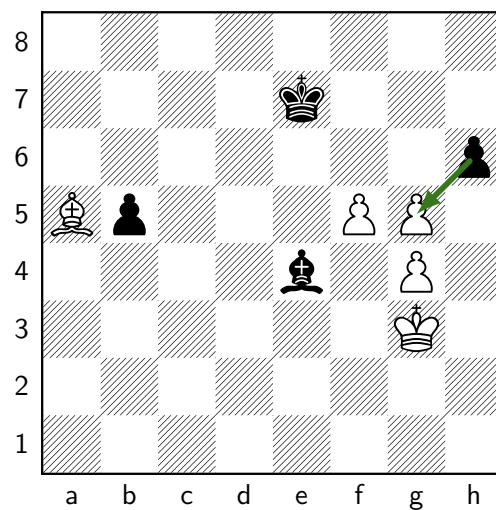


(d)

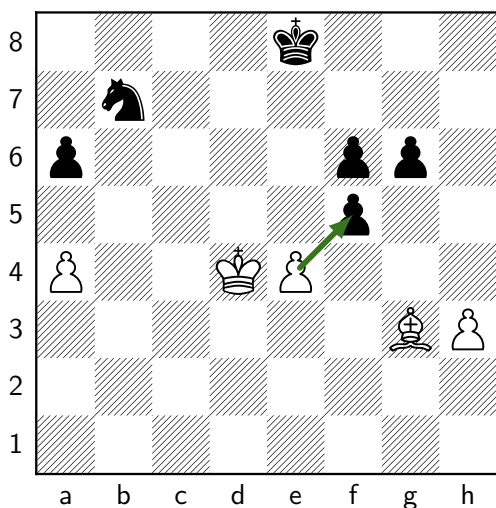
Figure 35: **Move 41, White: Bxe4.** (a) Our position; Melanie plays Bxe4 (f3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0003#157` (ply 93) [structural distance $d=27$, QAP obj=15]: the master played `gxf4`; under μ our *Bf3* plays the role of that game’s *Pg4* (a **cross-type** correspondence); cross-type roles: our *Pb2* as their *Bb7*, our *be4* as their *pf4*; transferred move `f3e4` (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#1859` (ply 94) [structural distance $d=23$, QAP obj=35]: the master played `Nf4`; under μ our *Nd7* plays the role of that game’s *Ng6*; cross-type roles: our *be4* as their *Pg2*, our *Pb2* as their *Bc4*; transferred move `d7c5`. (d) Retrieved from `ccr1_top.p0002#2033` (ply 77) [structural distance $d=24$, QAP obj=27]: the master played `bxc5`; under μ our *Pb2* plays the role of that game’s *Pb3*; cross-type roles: our *Bf3* as their *ph4*, our *Nd7* as their *Bc2*; transferred move `b2c3`. Synthesis: 1 retrieved analog transferred to Bxe4 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



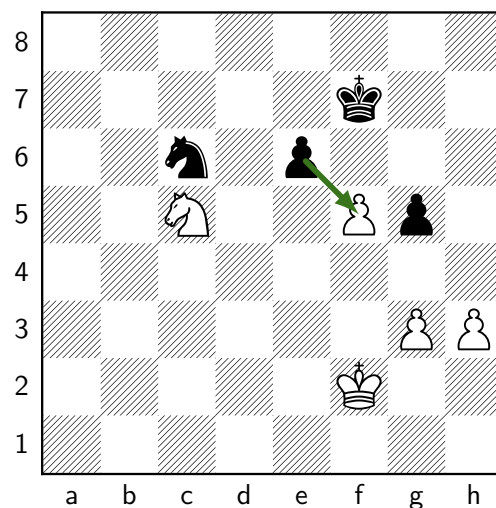
(a)



(b)

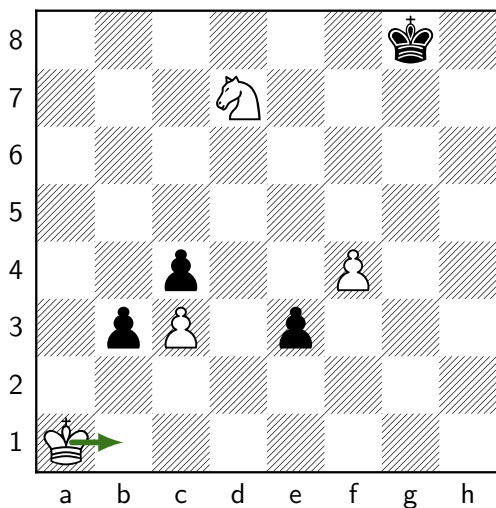


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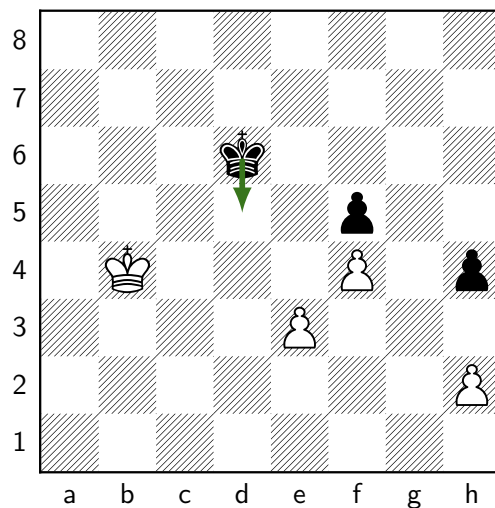


(d)

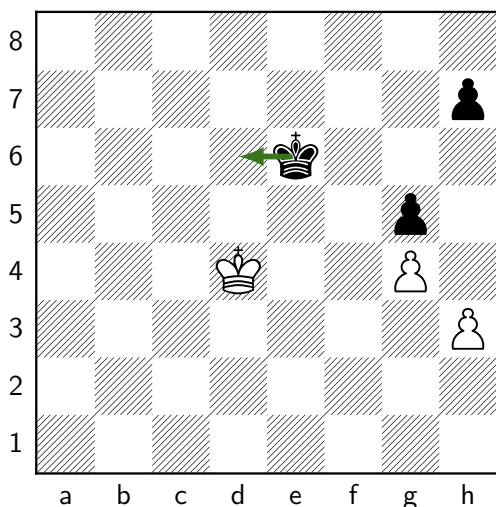
Figure 36: **Move 42, White: bxc3**. (a) Our position; Melanie plays **bxc3** (**b2c3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1211` (ply 69) [structural distance $d=15$, QAP obj=52]: the master played **hxg5**; under μ our *Pb2* plays the role of that game’s *Ph3*; cross-type roles: our *pb3* as their *ba5*, our *Pb2* as their *Be4*; transferred move **b2c3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3547` (ply 98) [structural distance $d=15$, QAP obj=46]: the master played **exf5**; under μ our *Pb2* plays the role of that game’s *Pe4*; transferred move **b2c3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2649` (ply 85) [structural distance $d=10$, QAP obj=44]: the master played **exf5**; under μ our *Pb2* plays the role of that game’s *Pe3*; cross-type roles: our *Pf4* as their *Nc6*, our *pc3* as their *nc5*; transferred move **b2c3** (**endorses this move**). Synthesis: 40 retrieved analogs transferred to **bxc3** (summed QAP objective 1552, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



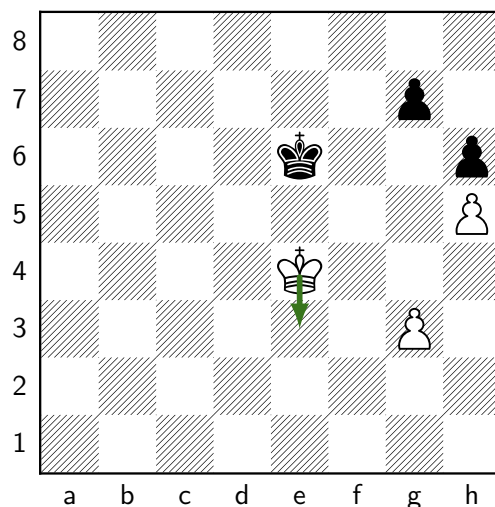
(a)



(b)

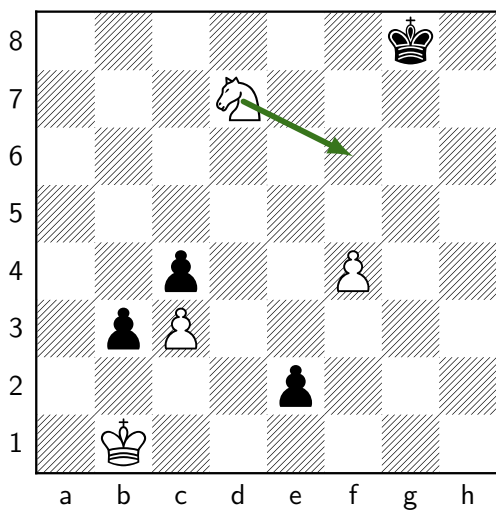


(c)

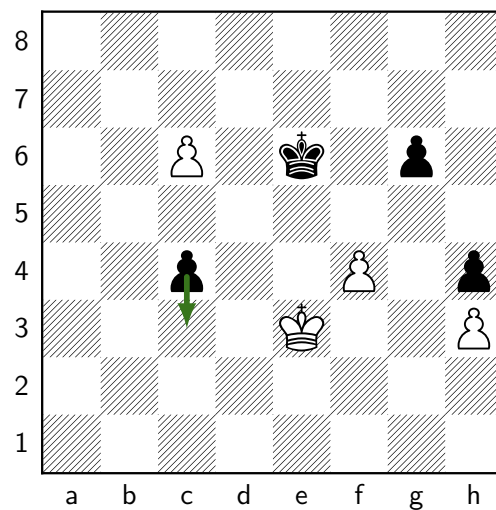


(d)

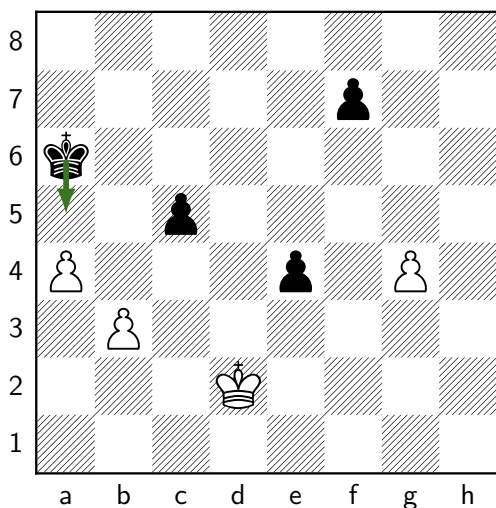
Figure 37: **Move 43, White: Kb1.** (a) Our position; Melanie plays Kb1 (**a1b1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3836` (ply 93) [structural distance $d=4$, QAP obj=2]: the master played Kd5; under μ our *Ka1* plays the role of that game's *Kd3*; transferred move **a1b1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#134` (ply 89) [structural distance $d=7$, QAP obj=2]: the master played Kd6; under μ our *Ka1* plays the role of that game's *Ke3*; transferred move **a1b1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1673` (ply 92) [structural distance $d=7$, QAP obj=2]: the master played Ke3; under μ our *Ka1* plays the role of that game's *Ke4*; transferred move **a1b1** (**endorses this move**). Synthesis: 21 retrieved analogs transferred to Kb1 (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



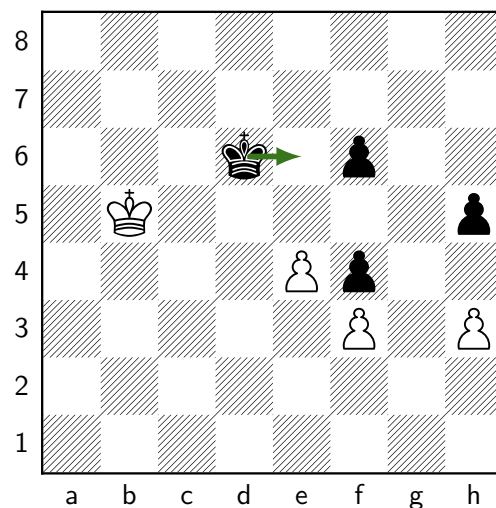
(a)



(b)

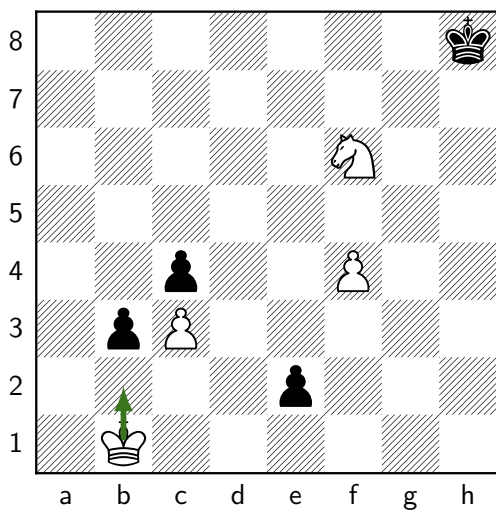


(c)

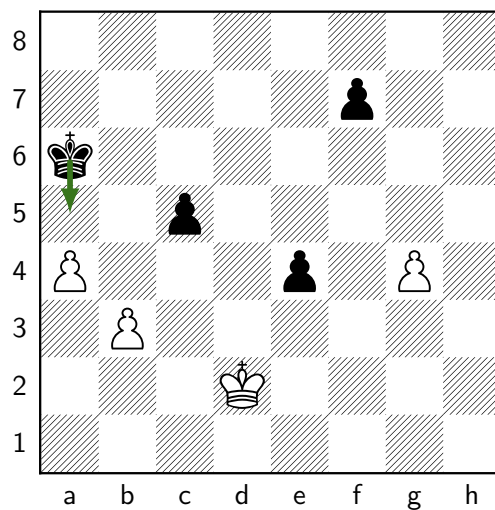


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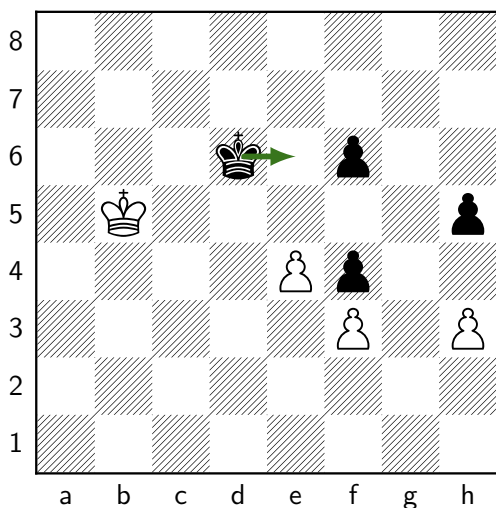
Figure 38: **Move 44, White: Nf6+ (d7f6).** (a) Our position; Melanie plays Nf6+ (d7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2624` (ply 97) [structural distance $d=7$, QAP obj=1]: the master played c3; under μ our Nd7 plays the role of that game’s Pc5 (a **cross-type** correspondence); transferred move d7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3260` (ply 89) [structural distance $d=3$, QAP obj=2]: the master played Ka5; under μ our Kb1 plays the role of that game’s Ka3; cross-type roles: our Nd7 as their pg4; transferred move b1b2. (d) Retrieved from `ccr1_top_p0003#2706` (ply 79) [structural distance $d=7$, QAP obj=2]: the master played Ke6; under μ our Kb1 plays the role of that game’s Kd3; cross-type roles: our Nd7 as their Pf4; transferred move b1b2. Synthesis: 1 retrieved analog transferred to Nf6+ (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



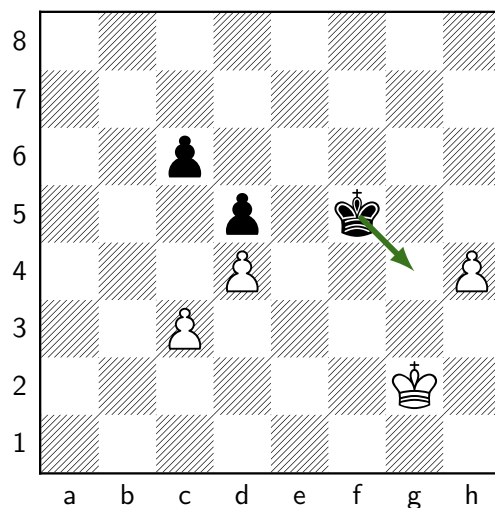
(a)



(b)

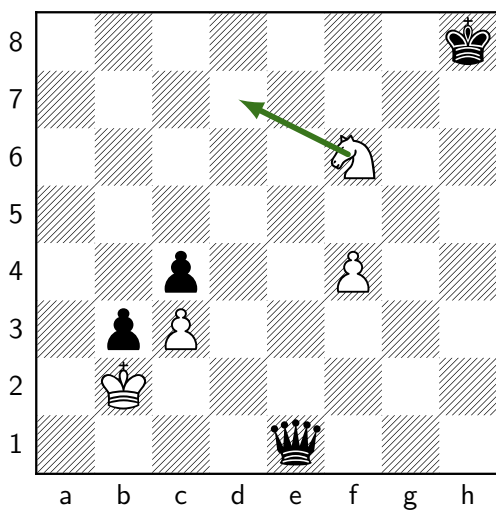


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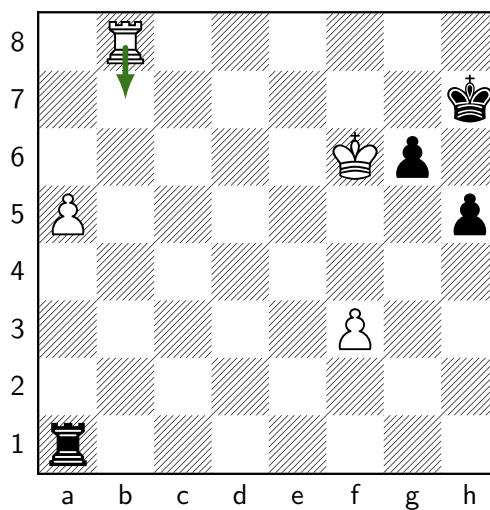


(d)

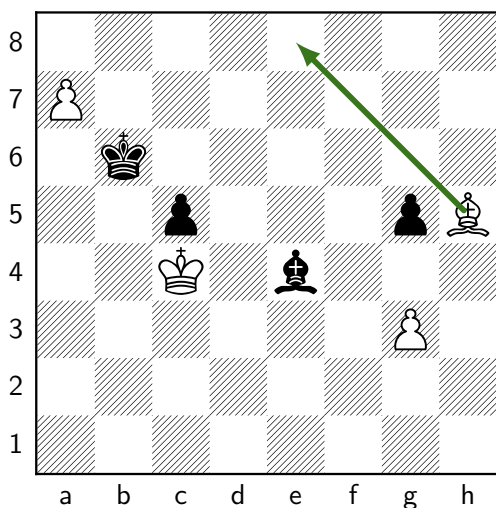
Figure 39: **Move 45, White: Kb2.** (a) Our position; Melanie plays Kb2 (b1b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3260` (ply 89) [structural distance $d=4$, QAP obj=2]: the master played Ka5; under μ our *Kb1* plays the role of that game’s *Ka3*; cross-type roles: our *Nf6* as their *Pe4*; transferred move b1b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2706` (ply 79) [structural distance $d=8$, QAP obj=2]: the master played Ke6; under μ our *Kb1* plays the role of that game’s *Kd3*; cross-type roles: our *Nf6* as their *Ph5*; transferred move b1b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#508` (ply 99) [structural distance $d=8$, QAP obj=2]: the master played Kg4; under μ our *Kb1* plays the role of that game’s *Kf4*; transferred move b1b2 (**endorses this move**). Synthesis: 15 retrieved analogs transferred to Kb2 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



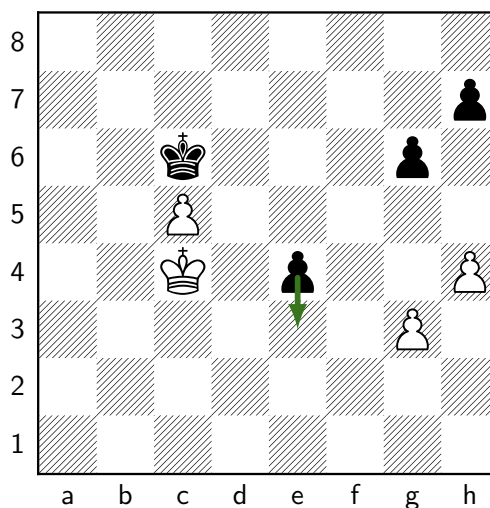
(a)



(b)

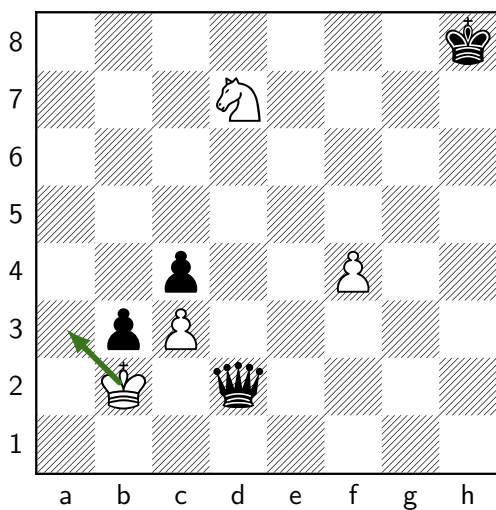


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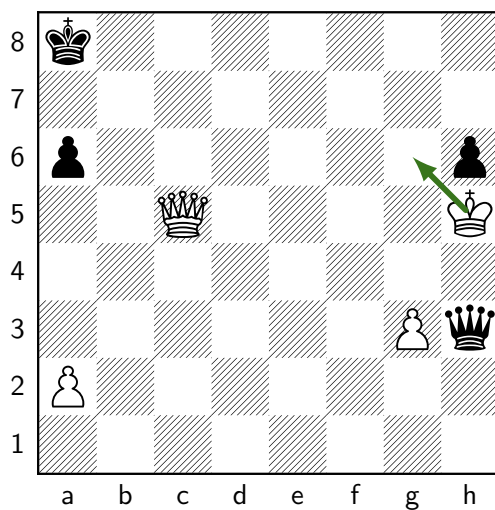


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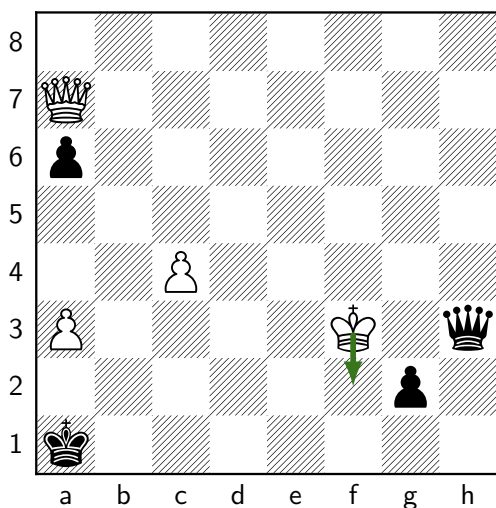
Figure 40: **Move 46, White: Nd7 (f6d7).** (a) Our position; Melanie plays Nd7 (f6d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#116` (ply 104) [structural distance $d=15$, QAP obj=23]: the master played Rb7+; under μ our Nf6 plays the role of that game’s Rb8 (a **cross-type** correspondence); cross-type roles: our qe1 as their ra1; transferred move f6d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2855` (ply 104) [structural distance $d=14$, QAP obj=19]: the master played Be8; under μ our Nf6 plays the role of that game’s Bh5 (a **cross-type** correspondence); cross-type roles: our qe1 as their be4; transferred move f6d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#455` (ply 89) [structural distance $d=15$, QAP obj=19]: the master played e3; under μ our Nf6 plays the role of that game’s Pe5 (a **cross-type** correspondence); cross-type roles: our qe1 as their ph4; transferred move f6d7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd7 (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



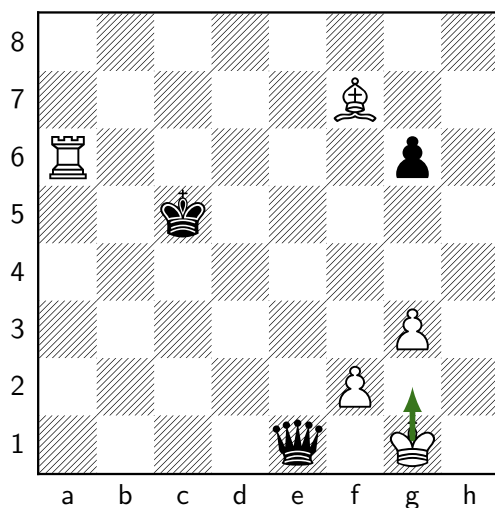
(a)



(b)

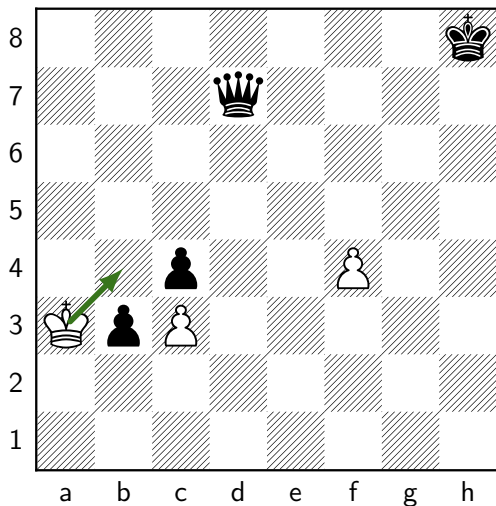


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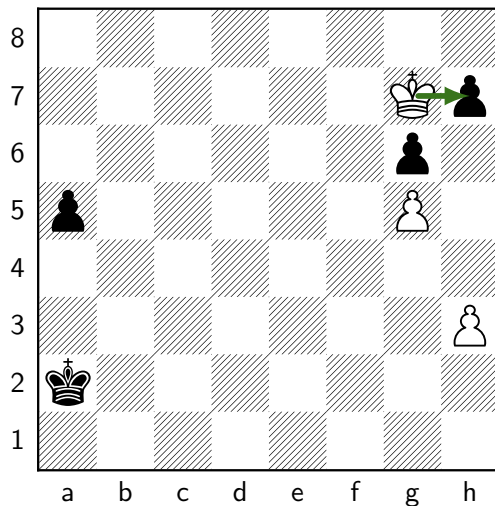


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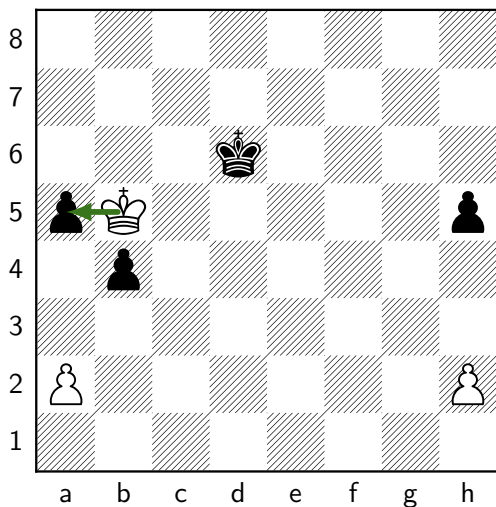
Figure 41: **Move 47, White: Ka3.** (a) Our position; Melanie plays **Ka3 (b2a3)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2049` (ply 86) [structural distance $d=29$, QAP obj=79]: the master played **Kg6**; under μ our *Kb2* plays the role of that game’s *Kh5*; cross-type roles: our *Nd7* as their *Qc5*; transferred move **b2a3 (endorses this move)**. (c) Retrieved from `ccrl_top_p0000#2798` (ply 92) [structural distance $d=34$, QAP obj=74]: the master played **Kf2**; under μ our *Kb2* plays the role of that game’s *Kf3*; cross-type roles: our *Nd7* as their *Pa3*, our *Pf4* as their *Qa7*; transferred move **b2a3 (endorses this move)**. (d) Retrieved from `ccrl_top_p0002#1082` (ply 88) [structural distance $d=30$, QAP obj=69]: the master played **Kg2**; under μ our *Kb2* plays the role of that game’s *Kg1*; cross-type roles: our *Nd7* as their *Ra6*, our *Pf4* as their *Bf7*; transferred move **b2a3 (endorses this move)**. Synthesis: 33 retrieved analogs transferred to **Ka3** (summed QAP objective 1613, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



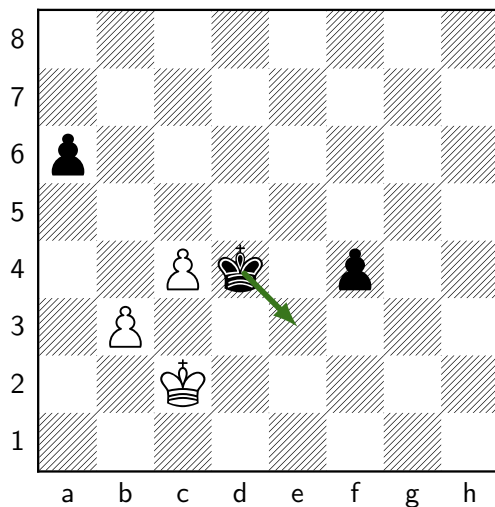
(a)



(b)

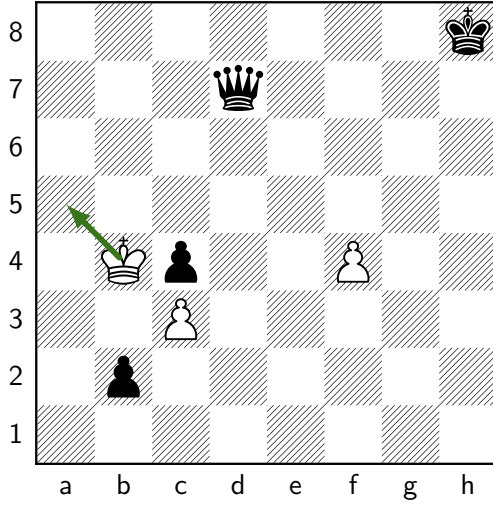


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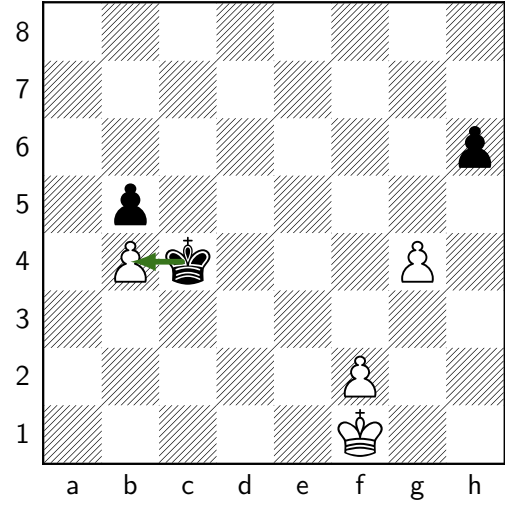


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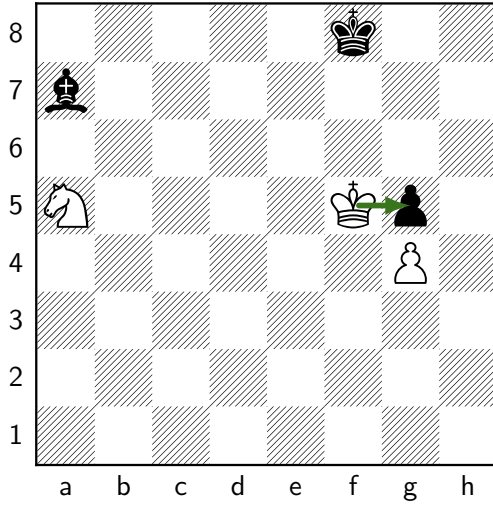
Figure 42: **Move 48, White: Kb4.** (a) Our position; Melanie plays Kb4 (a3b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2388` (ply 82) [structural distance $d=7$, QAP obj=32]: the master played Kxh7; under μ our $Ka3$ plays the role of that game’s $Kg7$; transferred move a3b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1048` (ply 88) [structural distance $d=6$, QAP obj=32]: the master played Kxa5; under μ our $Ka3$ plays the role of that game’s $Kb5$; transferred move a3b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1170` (ply 109) [structural distance $d=7$, QAP obj=21]: the master played Ke3; under μ our $Ka3$ plays the role of that game’s $Kd5$; transferred move a3b4 (**endorses this move**). Synthesis: 40 retrieved analogs transferred to Kb4 (summed QAP objective 792, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



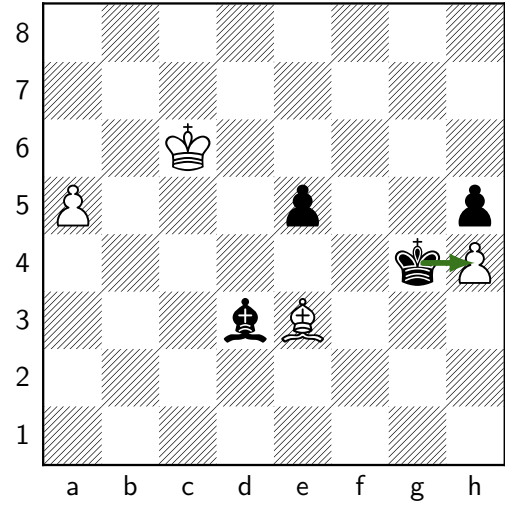
(a)



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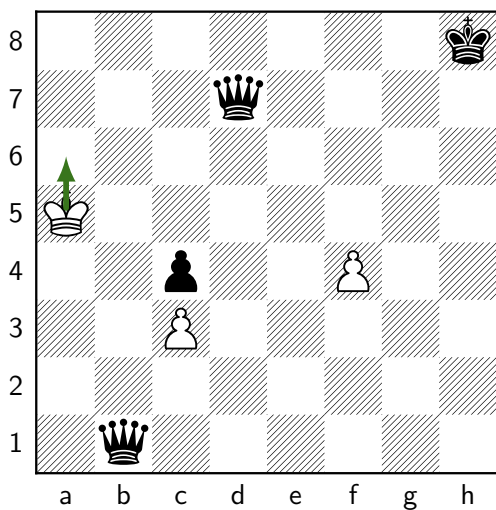


(c)

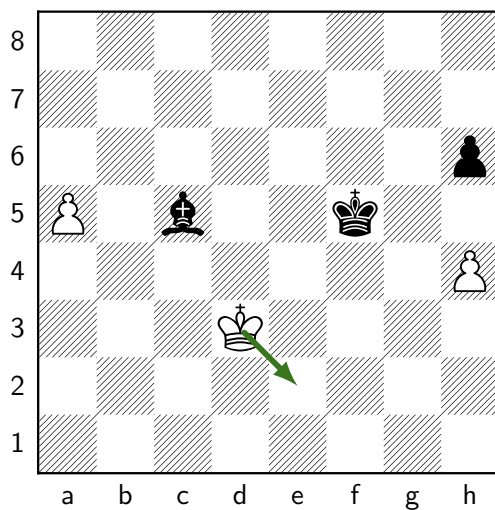


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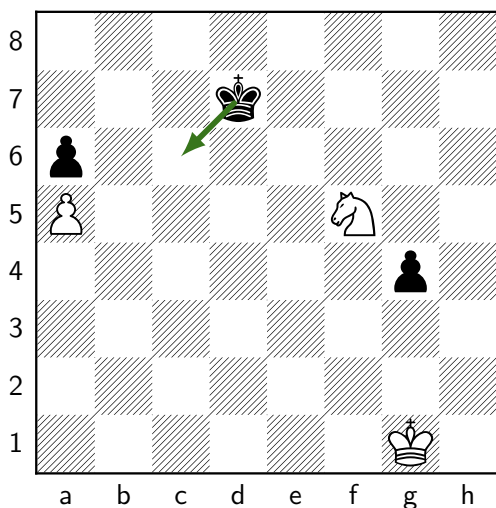
Figure 43: **Move 49, White: Ka5.** (a) Our position; Melanie plays Ka5 (b4a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0003#674` (ply 89) [structural distance $d=8$, QAP obj=24]: the master played Kxb4; under μ our Kb4 plays the role of that game’s Kc5; cross-type roles: our qd7 as their pg4; transferred move b4a5 (**endorses this move**). (c) Retrieved from `cctl_top_p0002#2686` (ply 90) [structural distance $d=7$, QAP obj=24]: the master played Kxg5; under μ our Kb4 plays the role of that game’s Kf5; cross-type roles: our Pf4 as their Na5, our pb2 as their ba7; transferred move b4a5 (**endorses this move**). (d) Retrieved from `cctl_top_p0003#1882` (ply 101) [structural distance $d=8$, QAP obj=24]: the master played Kxh4; under μ our Kb4 plays the role of that game’s Kg5; cross-type roles: our pb2 as their Bd3, our qd7 as their be3; transferred move b4a5 (**endorses this move**). Synthesis: 21 retrieved analogs transferred to Ka5 (summed QAP objective 482, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



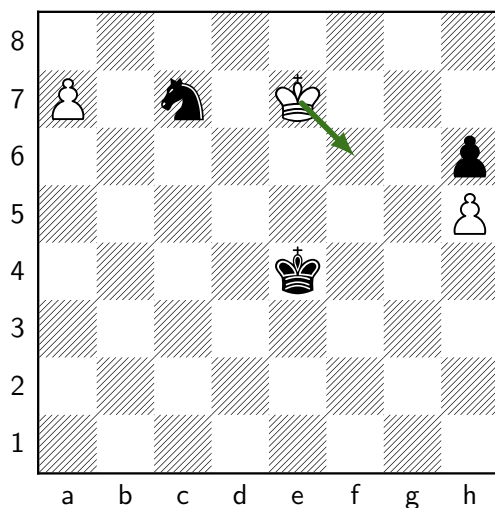
(a)



(b)



(c)



(d)

Figure 44: **Move 50, White: Ka6.** (a) Our position; Melanie plays Ka6 (a5a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3173` (ply 114) [structural distance $d=13$, QAP obj=1]: the master played Ke2; under μ our Ka5 plays the role of that game's Kd3; cross-type roles: our qb1 as their bc5; transferred move a5a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#632` (ply 105) [structural distance $d=13$, QAP obj=1]: the master played Kc6; under μ our Ka5 plays the role of that game's Kd2; cross-type roles: our qb1 as their nf5; transferred move a5a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3818` (ply 106) [structural distance $d=13$, QAP obj=1]: the master played Kf6; under μ our Ka5 plays the role of that game's Ke7; cross-type roles: our qb1 as their nc7; transferred move a5a6 (**endorses this move**). Synthesis: 39 retrieved analogs transferred to Ka6 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.